SWE5006 Practice Project · Graduate Certificate in Designing Modern Software Systems

# Speaker Script and Live Demo Run Book

## BlocBuy — Team 8 · Monday 26 October 2026, 16:30–17:15 SGT (Zoom)

| | |
|---|---|
| **Responsible** | Yu Qianshuo (Yorik), presentation lead and storyline (#464); Lim Jiechao, slide master. |
| **Companion files** | `swe5006-presentation.html` — the deck (22 slides + backups BX, B1–B16; open in Chrome or Edge; **S** notes window with this script and a timer, **T** start timer, **O** overview, **F** fullscreen, **B** backup index) · `swe5006-presentation.pdf` — the same deck as a PDF for submission (39 pages: 22 main, then BX and B1–B16). |
| **Single source** | `src/content.js` holds wording, timings, cues, Q&A, backups and open items; `src/slides-main.js` and `src/slides-backup.js` hold the slide layouts and `src/script.js` this document; shared UML diagrams come from `docs/presentation-kit/diagrams/`. Run `npm ci` once in `docs/presentation-kit/`, then `npm --prefix docs/swe5006/presentation run all` from the repository root. |
| **Facts as of** | 19 Oct 2026. Dev health `UP`; #223 and #517 open, so no payment has been confirmed and no refund proven on dev yet. Everything that depends on the 23 Oct go/no-go or the 24 Oct dry run is marked (§8). |

### How to use this document

- **Speakers:** read your segment (§1–§4) aloud against a timer at least twice. Each slide card shows the slot and the estimated speaking time at 130 words a minute; cards marked **over by N s** need a faster read or the listed trim. Rephrase in your own words; never read the slide.
- **Pipeline demo (Pan, Lim):** §4 after slide 20: preconditions, screen layout, then steps P1–P7 with what to operate, say and expect, and when to switch to the recording.
- **System demo (Eric, Yorik):** §5: the go/no-go table, the 16:05 expiry bloc, preconditions, screen layout, steps 1–8. §6 adds Plan C and time recovery.
- **Lead and slide master:** the run sheet below (interruption policy, background actions, coverage, cue words), §6, the day timeline in §7 and the open items in §8.

### Markers

`[FILL]` value that does not exist yet; never present a guess · `[VERIFY]` claim to check against the repository, report or run logs · `[CONFIRM]` needs team agreement. They are visible on the slides on purpose and must all be gone after the 25 Oct rehearsal.

Structure follows the lecturer's presentation sequence: one chapter per segment (1 Project introduction and management, 2 Analysis and design, 3 Testing, 4 CI/CD pipeline demo, 5 System demo), each opening with the guideline's own items, copied verbatim.

# Run sheet and coverage check

## Slide-by-slide run sheet

| Clock | Slot | Slide | Title | Speaker | Item | Est. |
|---|---|---|---|---|---|---|
| 16:30 | 15 s | 1 | BlocBuy | Lim | 1a | 13 s |
| 16:30:15 | 40 s | 2 | Group buying only works if enough people join in time | Lim | 1a | 37 s |
| 16:30:55 | 45 s | 3 | Scope: nine use cases on one platform | Lim | 1b | 44 s |
| 16:31:40 | 50 s | 4 | Four sprints, one board, one Definition of Done | Lim | 1e | 49 s |
| 16:32:30 | 50 s | 5 | Milestones and effort per member | Lim | 1d | 47 s |
| 16:33:20 | 40 s | 6 | Status, open issues, and what each retro changed | Lim | 1c | 38 s |
| 16:34 | 75 s | 7 | Top-down: split by transaction, one codebase, two topologies | Eric | 2a | 70 s |
| 16:35:15 | 45 s | 8 | UC9 listings: State, Optimistic Offline Lock, and the schema | Liu Jin | 2e | 42 s |
| 16:36 | 45 s | 9 | UC3/UC4 start and join: Factory Method and shared guards | Eric | 2e | 44 s |
| 16:36:45 | 45 s | 10 | UC6 expiry refund: never refund twice | Yorik | 2e | 44 s |
| 16:37:30 | 45 s | 11 | Notifications: Observer, Outbox, Idempotent Receiver | Pan | 2e | 44 s |
| 16:38:15 | 45 s | 12 | UC8 My blocs: the URL owns the view state | Lim | 2e | 43 s |
| 16:39 | 90 s | 13 | Seven kinds of tests, from domain rules to the deployed site | Liu Jin | 3a | 71 s |
| 16:40:30 | 90 s | 14 | Gates that block a merge, and what tests caught | Liu Jin | 3b | 77 s |
| 16:42 | 45 s | 15 | Source code and artefacts: one trunk, images by digest | Pan | 4a | 45 s |
| 16:42:45 | 60 s | 16 | One pipeline from pull request to dev | Pan | 4b | 56 s |
| 16:43:45 | 45 s | 17 | Security in the pipeline, no stored keys | Pan | 4d | 42 s |
| 16:44:30 | 45 s | 18 | Where it runs: dev on EC2, prod on ECS | Pan | 4b | 42 s |
| 16:45:15 | 45 s | 19 | Technical findings and issues | Pan | 4c | 41 s |
| 16:46 | 15 s | 20 | Live: one change from pull request to dev | Pan | 4b | 14 s |
| 16:53 | 20 s | 21 | Live demo: one bloc from invite to refund | Eric | 5a | 20 s |
| 17:07 | Q&A | 22 | Thank you | Lim | 5b | |

Segment talk budget vs slots vs estimated speaking time: §1 240 s / 240 s / 228 s · §2 300 s / 300 s / 287 s · §3 180 s / 180 s / 148 s · §4 660 s / 255 s / 240 s · §5 840 s / 20 s / 20 s (§5 talk is the demo itself).

Questions: during the talk (lecturers may ask at any time) · buffer **17:07–17:10** for questions asked during the talk · Q&A **17:10–17:15**. Segment 4 slides end with slide 20 at 16:46:15; the live pipeline demo fills 16:46:15–16:53:00 (§4). Segment 5 is slide 21 plus the system demo (§5).

**Interruption policy**

- A lecturer may ask at any moment. The current speaker stops, answers in **≤ 30 s** (decision → reason → evidence or limitation), says "back to …" and continues from the same slide. A short honest answer beats a long one.
- If the right owner is someone else, the speaker says "*Yorik owns that; he'll answer it at the end*" and Lim posts `Q: <question> → Yorik` in the backchannel. Parked questions are answered first in the buffer

(17:07–17:10), then in Q&A.
- The **3-minute buffer** pays for questions during the talk. Lim tracks the clock against the run sheet and posts `BEHIND n` when a segment starts late; the next speaker applies the cuts in §6. Hard checkpoints: 16:34 Eric, 16:39 Liu Jin, 16:42 Pan, 16:53 Eric.
- During a live demo the operator keeps operating while the narrator answers; a live step is never restarted because of a question.

## Background actions (off screen)

| Time | Action | Who |
|---|---|---|
| ≈ 16:05 | Yorik, as the leader account, starts the expiry bloc on `US34-04` (team of 2, 40 min) and pays with the Sandbox test card. It must expire ≈ 16:45 to show its refund by 16:53. Latest start 16:15. | Yorik |
| 16:10 | `EXPIRY OK` or `EXPIRY LATE` (bloc open at 1 of 2 on /account). | Yorik |
| 16:41 | During slide 14, muted: Pan runs `gh pr create` for the prepared demo branch (A.5) and posts `PR OPEN`. Checks take ≈ 4.8 min. | Pan |
| 16:46:15 | Share switch Lim → Pan (live pipeline demo). | Lim, Pan |
| ≈ 16:47:15 | `APPROVE`: Lim approves the demo pull request from his own machine. | Pan, Lim |
| 16:50 | `REFUND OK` or `REFUND LATE`. | Yorik |
| 16:53:00 | Share switch Pan → Eric (system demo, deck tab at slide 21). Eric keeps sharing to the end (slide 22, backups). | Pan, Eric |

## Every guideline item has slides and a speaker

| Item | Guideline requirement (verbatim) | Slides and demo steps | Speaker |
|---|---|---|---|
| **1a** | Explain the context and the business problem that the solution as a whole is trying to achieve. | 1, 2 | Lim |
| **1b** | Explain the scope of the project. | 3 (use case diagram) | Lim |
| **1c** | Explain the current status of the project including outstanding issues. | 6 | Lim |
| **1d** | Describe the actual project milestones and the total effort expended by each team member. | 5; B11 | Lim |
| **1e** | Explain how many sprints planned, Definition of Done (DoD) achieved. | 4 | Lim |
| **1f** | Sprint retrospective & reflection in the next sprint | 6 (what each retro changed) | Lim |
| **1g** | Agile process artifacts like product backlog, user stories, sprint plans, burndown charts, sprint reviews and sprint retrospectives. | 4 (artefacts named in the cues); B11 | Lim |
| **2a** | Software architectural decisions that influence the logical architecture. | 7; B8, B9 | Eric |
| **2b** | Transition from analysis to design | 7 (bottom strip); B7 | Eric |
| **2c** | Use case diagram | 3, referred to on 7 | Lim, Eric |
| **2d** | Analysis and design models | 7–12; B1, B2, B3, B4, B7 | Eric, Liu Jin, Yorik, Pan, Lim |
| **2e** | Design problems and patterns | 8, 9, 10, 11, 12; B6 | Liu Jin, Eric, Yorik, Pan, |

| | | | |
|---|---|---|---|
| | | | Lim |
| **2f** | Database schemas | 8 (right); B5 | Liu Jin |
| **3a** | Types of Testing Conducted | 13; B10 | Liu Jin |
| **3b** | Well structured source code with unit test cases for classes with more complex behaviours. | 13, 14; B10 | Liu Jin |
| **4a** | Describe the strategy to manage the source code and artifacts. | 15 | Pan |
| **4b** | Describe the DevSecOps pipeline and procedures. | 16, 18, 20 + live pipeline demo (steps P1–P7); B12 | Pan |
| **4c** | Highlight any technical findings and issues encountered. | 19, 14 | Pan |
| **4d** | Design specification of the DevOps pipelines along with the use of appropriate tools with minimal security considerations. | 16, 17; B14 | Pan |
| **4e** | Backup: Record a video in advance in case of any technical issues during the live demo. | 20 (fallback box) + the 25 Oct recording | Pan |
| **5a** | Demonstrate some of the significant use cases of the solution. | 21 + demo steps 1–4 | Eric, Yorik |
| **5b** | Demonstrate how the solution deals with some of the important quality attributes e.g. scalability (such as demonstrating auto-scaling), performance (show using some scripted performance testing), maintainability (show/run the DevOps pipeline). | demo steps 5–7; B4, B15, B16; slides 9, 10, 15–20 | Yorik, Eric |
| **5c** | Backup: Record a video in advance in case of any technical issues during the live demo. | the 25 Oct recordings, one clip per step | Yorik, Eric |

## Presentation tips checklist

| Tip | How this deck meets it |
|---|---|
| Summarize and highlight the significant achievements. Talk more about the significant and interesting parts of the solution and implementation. Your audience should remember the unique aspects of the solution and not similarities to other solutions. | The unique parts get the time: pay-first teams that refund themselves (slides 2, 10), one codebase in two topologies (slide 7), money safety under crashes and races (slides 9, 10), a pipeline from PR to dev in minutes (slides 16, 20). Slide 22 repeats them. |
| Manage your time well. Do not present too many slides. Remember that the audience can read the text in the slide so refrain from reading them. | 22 main slides in 37 minutes, two of them live demos; slots checked at 130 wpm in the table above; the sample wording adds to the slide instead of reading it; details sit in backups B1–B16. |
| Explain your design in top-down manner. Make sure that your audience understands the requirements, the high-level design and the high-level decisions that you make before drilling into some details. | Requirements and scope first (slides 2–3), then the top-down architecture slide (7), then one use case and design problem per member (8–12). |
| Ensure that the contents on the slides are readable from the distance at which your audience is sitting. This is particular true for face-to-face presentation. | On Zoom: body text ≥ 15 px on a 1280 × 720 canvas, diagram labels ≥ 14.5 px (STYLE.md); one message per slide title. |
| Lecturers may ask questions during the presentation, please be prepared to respond accordingly and continue the presentation. (lecturer's note) | Interruption policy above: answer in ≤ 30 s, then continue; the 3-minute buffer 17:07–17:10 absorbs them; Lim parks longer ones. |
| Backup: Record a video in advance in case of any technical issues during the live demo. (lecturer's note; items 4e and 5c) | Both demos recorded on dev by 12:00 on Sun 25 Oct; one clip per step, cued with `CLIP n` (§4, §5). |

## Cue words in the team backchannel (never the Zoom chat)

| Message | Meaning | Sent by |
|---|---|---|
| `1 MIN` | One minute of talk left in this segment | Lim |
| `TIME` | Stop; finish the sentence and hand over | Lim |
| `Q: <text> → name` | A question was parked; the named owner answers it in the buffer or in Q&A | Lim |
| `BEHIND n / AHEAD n` | The talk is n seconds behind or ahead of the run sheet; the next speaker applies the §6 cuts | Lim |
| `EXPIRY OK / EXPIRY LATE` | 16:10: the 16:05 bloc is open at 1 of 2 on /account, or it is not (retry rule in §6) | Yorik |
| `PR OPEN` | Pan has opened the demo pull request (target 16:41) | Pan |
| `APPROVE` | Lim approves the demo pull request now, from his own GitHub session | Pan → Lim |
| `REFUND OK / REFUND LATE` | 16:50: the receipt of the 16:05 payment shows the refund, or it does not yet | Yorik |
| `SHARE` | Next person: start screen sharing now (16:46:15 Lim → Pan; 16:53:00 Pan → Eric) | Lim |
| `CLIP n` | Live step n failed; play the recording of step n (pipeline steps use `CLIP Pn`) | Pan or Eric |

# 1 Project Introduction & Project Management Summary (4+0 minutes)

| | |
|---|---|
| **Clock** | 16:30:00 – 16:34:00 (4 min talk + 0 min Q&A; talk ends 16:34:00) |
| **Owner** | Lim Jiechao |
| **Slides** | 1, 2, 3, 4, 5, 6 |

**Lecturer's presentation sequence · Presentation Guidelines · Briefing deliverables (V1.0)**

**1a** — Explain the context and the business problem that the solution as a whole is trying to achieve.
**1b** — Explain the scope of the project.
**1c** — Explain the current status of the project including outstanding issues.
**1d** — Describe the actual project milestones and the total effort expended by each team member.
**1e** — Explain how many sprints planned, Definition of Done (DoD) achieved.
**1f** — Sprint retrospective & reflection in the next sprint
**1g** — Agile process artifacts like product backlog, user stories, sprint plans, burndown charts, sprint reviews and sprint retrospectives.

**Tip**
Lim opens at 16:30:00 sharp, with the deck already shared and the notes window off the shared screen. Slide 6 carries the go/no-go: after 23 Oct, say in one sentence whether the payment steps of the demo are live or recorded, so nobody is surprised at 16:53.

## Slide 1 BlocBuy

16:30:00–16:30:15 · 15 s

**Speaker** Lim · covers guideline item **1a**

**Say**
*Good afternoon, we're Team 8, and this is BlocBuy. I'll cover the problem and how we ran the project; then come design, testing, our pipeline, and a live demo.*
≈ 29 words · ≈ 13 s at 130 wpm · slot 15 s

**Cues**
- 16:30:00 — Lim opens swe5006-presentation.html in Chrome, presses S for the notes window (keep it off the shared screen) and T there to start the timer. In Zoom, share only the deck window.
- Backchannel: the team chat stays open during the talk. Anyone who sees the share or audio drop posts there; Lim repeats any question asked during the talk in one line and names who answers (questions may come at any time, Q&A is 17:10–17:15).
- Tagline source: proposal §4 (two prices, pay first, automatic refund).

**Hand-over** *"So, what problem are we solving?"*

## Slide 2 Group buying only works if enough people join in time

16:30:15–16:30:55 · 40 s

**Speaker** Lim · covers guideline item **1a**

**Say**

*Small merchants can only offer a discount when volume is predictable, and a single buyer can never reach a bulk price. Advertising tries to bridge that gap, but it's expensive and not tied to real purchases. BlocBuy solves this with pay-first teams. Every listing has two prices. You pay upfront to start a team and share an invite link; friends join by paying too. If the team fills in time, everyone gets the team price. If not, everyone is refunded automatically.*

≈ 81 words · ≈ 37 s at 130 wpm · slot 40 s · if short of time: If over time, drop "Advertising tries to bridge that gap... real purchases."

**Cues**

- Source: SWE5006 proposal §4 (27 Aug; the pay-first mechanic, two prices, the early Pinduoduo playbook) and §6.1 (Stripe test mode only). The README describes generic organisers and sellers; the proposal is the scope we built, so the slide follows it.
- What is unique: money is committed before the team exists, so every buyer has a reason to recruit, and the refund on expiry makes joining risk-free.

**Hand-over** *"Here's what we built to do that."*

## Slide 3 Scope: nine use cases on one platform

16:30:55–16:31:40 · 45 s

**Speaker** Lim · covers guideline item **1b**

**Say**

*This is our scope as a use case diagram. Buyers browse listings, sign in with Google, start or join a team, track their orders and teams, and receive notifications. Starting and joining both include paying upfront through Stripe Checkout. Joining through an invite link credits the person who shared it. A scheduler resolves each team, and an expired team triggers automatic refunds through Stripe. Admins manage listings and group terms. The initials show who owns each use case in our report. Out of scope: a merchant self-service GUI, analytics, recommendations, email or push, and live payments.*

≈ 96 words · ≈ 44 s at 130 wpm · slot 45 s · if short of time: If over time, drop "The initials show who owns each use case in our report."

**Cues**

- Diagram: src/diagrams/use-case.svg. UC1–UC9 are the proposal's (§6.1); "Receive notifications" joined in Sprint 4 (#215, #446) and is Pan's report use case (#463). Out-of-scope list: proposal §6.2.
- «include»: START and JOIN share one Checkout path (CheckoutQuote accepts only START or JOIN, backend payment/model). «extend»: UC7 credits the sharer only when a join arrives through an invite link (proposal UC7); UC6 runs only when a team expires (proposal UC5, UC6).
- Initials = report owner per #463: LJ Liu Jin, WZ Eric, YQ Yorik, PS Pan, JC Lim. Beyond the proposal: the API-only admin (UC9) gained an admin console in Sprint 3 (PRs #305–#307, 30 Sep).

**Hand-over** *"And this is how we ran the project."*

## Slide 4 Four sprints, one board, one Definition of Done

16:31:40–16:32:30 · 50 s

**Speaker** Lim · covers guideline item **1e**

***Say***

*We planned four two-week sprints. Each sprint has a plan and a review, and from Sprint 3 we added a mid-sprint check; from Sprint 2, every working day has a dated log in the repository. The backlog is one GitHub Project board: 159 cards today, 132 of them done. We merged 356 pull requests, 70 of them Dependabot updates. We estimate in hours, 30 per member per sprint. Our Definition of Done grew after each retro: on dev means deployed, not local; pull requests stay under 1,500 lines; every quality gate blocks the merge. That strictness is why Sprint 3's goal does not count as met.*

≈ 106 words · ≈ 49 s at 130 wpm · slot 50 s · if short of time: If over time, drop "We estimate in hours, 30 per member per sprint." (it is on slide 5).

**Cues**

- Counted 19 Oct with gh: board "GroupBuy Kanban Board" (project 2) 159 cards = 132 Done, 7 In progress, 15 Ready, 5 Backlog; 159 issues = 134 closed (119 completed, 15 not planned), 25 open; 356 merged PRs = 263 team, 70 Dependabot, 23 board-snapshot PRs. Sprint windows by merge date (SGT), Monday to the Sunday before the next sprint; Sprint 1 includes 7 PRs from 24 Aug.
- Sprint calendar: docs/agile/README.md (the proposal's dates differ by up to three days). Goals: Sprint 1 report, Sprint 2 plan (docx), docs/agile/sprint-03/plan.md §2, sprint-04/plan.md §2. Results: Sprint 2 report (9 of 11 stories, 134 of 150 h), sprint-03/review.md (117 of 150 h, goal not met on dev).
- Artefacts (item 1g): backlog = board + issue templates (use case, task, NFR); user stories with acceptance criteria in each plan; sprint plans in docs/agile/sprint-0N/plan.md (Sprints 1–2 as docx); burndowns from docs/agile/metrics/ (backup B11); reviews and retros in the progress reports. DoD text: sprint-03/plan.md §8, sprint-04/plan.md §8.
- Point at "Was it met?": the gates are required checks; the estimate field in the 12–19 Oct snapshots still holds each card's planned hours, not remaining hours.

**Hand-over** *"Here's how that adds up over time."*

## Slide 5 Milestones and effort per member

16:32:30–16:33:20 · 50 s

**Speaker** Lim · covers guideline item **1d**

***Say*** *Our milestones: the proposal on 27 August, the backend live on dev on 5 September, the web app on 22 September, and a release tag after each of the first three sprints. Sprint 2 finished 134 of 150 planned hours, Sprint 3 117 of 150. Sprint 4 carries 224.5 hours across two courses; by the 19 October snapshot, cards worth 118 planned hours were done, which is the burndown on the right. On the left are planned hours per member. We did not log actual hours per person, so this is the plan, against a guideline of about ten man-days each.*

≈ 101 words · ≈ 47 s at 130 wpm · slot 50 s · if short of time: If over time, drop the milestone dates after 5 September.

**Cues**

- Sources: proposal (27 Aug); Sprint 1 report (dev live 5 Sep); Sprint 2 report (web app on dev 22 Sep; 150/134 h); tags v0.1.0-sprint1/2/3 (14 Sep, 27 Sep, 11 Oct; git tag); sprint-03/review.md (US34 listings on dev 8 Oct; 150 h, 33 h remaining at freeze); board: #455 and #456 Done in the 15 Oct snapshot.
- Burndown: docs/agile/metrics/2026-10-12…19.csv on the metrics/board-snapshots branch, the planned estimate of every Sprint 4 card not in Done (224.5 → 106.5 h); ideal line from sprint-04/plan.md §7. The 19 Oct file was captured early on 19 Oct, so it shows the state after the weekend.
- Per-member hours: Sprint 2 plan (30 h each), sprint-03/plan.md §4 (30 h each), sprint-04/plan.md §5 as planned (51, 48, 46, 41.5, 38).
  `[VERIFY: open PR #538 moves #468 and #458 after Planning: Liu Jin 66, Eric 26, Pan 46.5; use those once it merges]`
- `[CONFIRM: actual effort per member, agreed by all five; never present the planned hours as actuals]`
  Board items Done per assignee (19 Oct) count cards, not effort; a card shared by two people counts for both, and 21 Done cards have no assignee.

**Hand-over** *"So where does that leave us today?"*

## Slide 6 Status, open issues, and what each retro changed

16:33:20–16:34:00 · 40 s

**Speaker** Lim · covers guideline item **1c**

***Say*** *Where we stand on 19 October: the dev site is up, with the web app, sign-in and 14 listings. Stripe Checkout opens and returns, but the webhook never confirms the payment, so paid teams and refunds don't run on dev yet. That's issues 517 and 223, and our go/no-go for the live demo. Each retro changed the next sprint: no burndown in Sprint 1 led to daily board snapshots; Sprint 2's huge pull requests led to the 1,500-line rule and the mid-sprint check.*

≈ 83 words · ≈ 38 s at 130 wpm · slot 40 s · if short of time: If over time, drop the second retro example (Sprint 2).

**Cues**

- Checked 19 Oct 14:14 SGT: curl https://d1b3np88z657cp.cloudfront.net/api/actuator/health → {"status":"UP"}; / → 200; /api/v1/products → 14 listings. Sign-in cookie: sprint-03/review.md (NFR-01 verified on dev, 8 Oct). #517 and #223 open on 19 Oct (gh issue view). Re-check all of this on 25 Oct and update the left column.
- Retro sources: Sprint 1 report (retrospective), Sprint 2 report §Sprint Retrospective → sprint-03/plan.md §8 (three new DoD rules). Sprint 3 retro is owed in #451; Sprint 4 retro is on 23 Oct.

**Hand-over** *"Eric will take you through the design."*

## Watch out

Never say that payments or refunds work on dev unless the 24 Oct dry run proved it (§5 decision table). Re-check slide 6 on 25 Oct (open item 8). `[CONFIRM: slide 6 wording after the 23 Oct go/no-go]`

### Q&A (during the talk / 17:10–17:15)

| Likely question | Answer: decision → reason → evidence or limitation (30–45 s) | Owner |
|---|---|---|
| Why does Sprint 3 count as not met when the code was merged? | Our Definition of Done says "on dev" means shown on the deployed environment, and a local run does not count. Checkout and scheduling were still off on dev at the freeze (#223), so the review recorded the goal as not met and carried it into Sprint 4 as the critical path. Evidence: sprint-03/review.md. | Lim |
| Are these hours actuals? | No. We planned in hours per story; done hours are planned hours of finished work (Sprint 2 report, Sprint 3 review), and the Sprint 4 burndown sums planned estimates of cards not yet Done. Actual hours per person were not logged, which is a limitation we state. | Lim |
| Was the effort balanced across the five members? | The plans gave everyone 30 h in Sprints 2 and 3; Sprint 4 started at 38–51 h each and was rebalanced after Planning (PR #538). Board cards differ in size, for example one Stripe pull request was about 10,000 lines, so card counts are not effort. `[CONFIRM: the effort split all five agree on]` | Lim |
| How did the retrospectives change what you did? | Each retro produced a rule in the next plan: daily board snapshots after Sprint 1; after Sprint 2, PRs under 1,500 lines, a mid-sprint check and remaining hours on the board. The last one was only partly kept: the estimate field still holds planned hours. Sources: Sprint 2 report, sprint-03/plan.md §8. | Lim |
| What is your biggest open risk for the demo? | The Stripe webhook on dev (#517): payments stay waiting, so paid START, JOIN and refunds do not complete on dev yet. Go/no-go for the live payment demo is 23 Oct, with a fallback recording by 25 Oct. | Lim → Yorik |

# 2 Analysis & Design and Design Patterns (5+0 minutes)

| | |
|---|---|
| **Clock** | 16:34:00 – 16:39:00 (5 min talk + 0 min Q&A; talk ends 16:39:00) |
| **Owner** | Wang Ziang (Eric) with Liu Jin, Yu Qianshuo (Yorik), Panpakorn Siripanich (Pan), Lim Jiechao |
| **Slides** | 7, 8, 9, 10, 11, 12 |

**Lecturer's presentation sequence · Presentation Guidelines · Briefing deliverables (V1.0)**

**2a** — Software architectural decisions that influence the logical architecture.
**2b** — Transition from analysis to design
**2c** — Use case diagram
**2d** — Analysis and design models
**2e** — Design problems and patterns
**2f** — Database schemas

**Tip**

Five speakers in five minutes: each design slide is 45 s. The next speaker unmutes during the previous speaker's last sentence and gives their name once ("I'm Liu Jin."). Each speaker answers questions about their own slide; backups B1–B9 sit one key away (B).

## Slide 7 Top-down: split by transaction, one codebase, two topologies

16:34:00–16:35:15 · 75 s

**Speaker** Eric · covers guideline item **2a**

***Say***

*I'm Eric. We designed top-down, from the use cases on slide 3. First decision: a modular monolith. The code has five COLA modules, and each bounded context, identity, catalog, group buying, order, payment and notification, is a package in every module. Eighteen ArchUnit rules fail the build when a dependency points the wrong way. Second: split only where no transaction crosses. Paying, joining the team, creating the order and taking stock commit together, so those four contexts stay together as commerce. Third: one codebase, two topologies. For this course, dev runs the monolith on one EC2 host. For SWE5001, the same modules are assembled into three services on ECS, linked by an outbox and SNS and SQS. At the bottom is our rule from analysis to design: a context becomes a package, an entity becomes an aggregate with a repository port, and an external actor becomes a gateway port with an adapter.*

≈ 152 words · ≈ 70 s at 130 wpm · slot 75 s · if short of time: If over time, drop "Eighteen ArchUnit rules fail the build when a dependency points the wrong way." (it stays on the slide).

**Cues**

- Sources: backend/pom.xml (modules blocbuy-client, -adapter, -application, -domain, -infrastructure, -bootstrap; services/*-svc); backend/docs/design/backend-skeleton-plan.md (layer rules, gateway ports); ArchitectureTest.java (18 @ArchTest rules, e.g. domain_is_framework_free, commerce_reaches_identity_only_through_user_port, nothing_depends_on_notification); backend ADR 0002 (events, consequences before commit), 0007 (outbox, SNS, SQS), 0008 (two topologies, split follows transactional consistency).
- Point left (2a: modules and contexts), right (2a: topology), then the bottom strip (2b: transition rule). 2c: say "the nine use cases on slide 3"; do not go back to it. Worked transition examples per member: B7.
- Backups for questions: B1 context map, B2 aggregates, B9 logical view (ECS topology), B8 ADR index. ADR 0007 still reads "Proposed" in its file although it is merged and runs on ECS; say so if asked.

**Hand-over** *"Liu Jin takes the first use case."*

## Slide 8 UC9 listings: State, Optimistic Offline Lock, and the schema

16:35:15–16:36:00 · 45 s

**Speaker** Liu Jin · covers guideline item **2e**

***Say*** *I'm Liu Jin. Admins draft, publish and delist listings. Two problems: which moves are legal depends on the status, and two admins can edit one listing. For the first we use State: each ProductStatus constant overrides only the moves it allows; anything else is refused with a 409. For the second, Optimistic Offline Lock: every write sends the version the admin loaded, and the update matches only that version, so a stale edit gets a 412 and a Reload. On the right, our schema: 29 tables, grouped by owning context.*

≈ 90 words · ≈ 42 s at 130 wpm · slot 45 s · if short of time: If over time, drop "and two admins can edit one listing" from the problem sentence.

**Cues**

- `[VERIFY: Liu Jin to confirm against his §3.4/3.6 draft]` His draft is not in docs/swe5006/report-drafts/; the wording comes from the code.
- State: ProductStatus.java, enum constants override publish(), delist(), deletionError(), requireCommercialEdit(); the defaults throw CATALOG_409_002 ("State-pattern default"). Product.publish()/delist() call status.publish()/delist() and never branch on status. Tests: ProductStatusTest (onSalePublishThrowsStateConflict, offSaleDelistThrowsStateConflict).
- Lock: Product.requireVersion(expected) and BizProductMapper.xml "WHERE … AND version = #{expectedVersion}"; ProductRepositoryImpl.requireChanged turns 0 rows into CATALOG_412_002 with {currentVersion}. expectedVersion is a body field (a query parameter on DELETE), not If-Match. Frontend ADR 0015: 412 → "You loaded v4; the server now has v5" banner and Reload. Test: AdminListingFlowIT.staleListingWritesReturnStoredVersionWithoutMutation.
- Schema (2f): 28 Flyway files V1–V28 in blocbuy-bootstrap/src/main/resources/db/migration, 29 CREATE TABLEs (V1 is a placeholder with the conventions). Grouping by owner follows backend ADR 0008. Full list: B5.

**Hand-over** *"Eric, starting and joining a team."*

## Slide 9 UC3/UC4 start and join: Factory Method and shared guards

16:36:00–16:36:45 · 45 s

**Speaker** Eric · covers guideline item **2e**

***Say***

*A team is created only after its first buyer has paid. Creation is a Factory Method: PaidGroupDealCreator's final initiate method checks the terms and the price, then calls createGroupDeal, which the standard creator implements. Our design problem was the join rules. They were written twice, for the quote and after payment, and they drifted: a team with no members passed the quote and failed after payment, so the buyer paid and was refunded. We moved the three rules into the GroupDeal aggregate; both paths call them. Honestly: there is one creator, wired at build time.*

≈ 95 words · ≈ 44 s at 130 wpm · slot 45 s · if short of time: If over time, drop "Honestly: there is one creator, wired at build time." (it stays on the slide).

**Cues**

- `[VERIFY: PR #542 not merged; Eric confirms]` Draft: git show origin/docs/463-eric-wang-draft:docs/swe5006/report-drafts/eric-wang-contribution.md.
- Checked in code 19 Oct: PaidGroupDealCreator (abstract; public final initiate; protected abstract createGroupDeal), StandardPaidGroupDealCreator (GroupDeal.startPaid), GroupDealCreationConfig (@Bean, the only creator), PaidGroupDealService (final constructor-injected creator, #421); GroupDeal.requireJoinWindow, requireAvailableSeat, requireNewMember called by joinPaid and by GroupDealPurchaseSupportImpl.lockJoinTerms (#422). PR #424 merged 13 Oct.
- Tests: PaidGroupDealServiceTest.startUsesTheInjectedCreator; GroupDealPurchaseSupportImplTest.aTeamWithoutItsPaidLeaderCannotBeQuoted (counts 0 and -1). Issue #463 words it "Factory Method / policy objects": policy objects were rejected because clients need the distinct codes DEAL_CLOSED, DEAL_FULL, ALREADY_MEMBER.

**Hand-over** *"Yorik takes it from expiry."*

## Slide 10 UC6 expiry refund: never refund twice

16:36:45–16:37:30 · 45 s

**Speaker** Yorik · covers guideline item **2e**

***Say*** *I'm Yorik. When a team expires, everyone is refunded. Expiry publishes TeamExpired, and two observers cancel the orders and record one refund per payment before the same commit. The risk: Stripe accepts a refund, our process dies before saving, and a naive retry refunds twice. So every call carries an idempotency key built from the obligation ID; a retry gets the same refund back. A worker claims each refund under a lease, retries after a fixed 30 seconds, and hands mismatches or anything past 23 hours to a human. Proven locally, not yet on dev.*

≈ 95 words · ≈ 44 s at 130 wpm · slot 45 s · if short of time: If over time, drop "and hands mismatches or anything past 23 hours to a human" (it stays on the slide).

**Cues**

- Source: docs/swe5006/report-drafts/yorick-yu-contribution.md, simplified. Code: RefundWork.providerKey() = "blocbuy-refund-" + id; StripeRefundGateway .setIdempotencyKey(work.providerKey()); RefundMapper.xml: claim sets lease_until +2 minutes and version+1, retry sets next_attempt_at +30 seconds, finish matches version; RefundService: ≥ 23 h since the first attempt → search Stripe only, no match → review; RefundWorkerJob every PT30S.
- Issue #463 says "Retry with Backoff"; the code is a fixed 30 s delay. Say "fixed delay", never "backoff". Observer: OrderLifecycleSubscriber @Order(10) and TeamExpiryRefundSubscriber @Order(20), both BEFORE_COMMIT (backend ADR 0002).
- Evidence limits: backend/docs/payment/stripe-refund-recovery-acceptance-20261003.md (local PostgreSQL, provider substitute); RefundServiceTest.lostResponseRetriesSameDurableKey. No refund on dev yet (#223, #517). Candidates and decisions: B6; sequence: B4.

**Hand-over** *"Pan, the notifications."*

## Slide 11 Notifications: Observer, Outbox, Idempotent Receiver

16:37:30–16:38:15 · 45 s

**Speaker** Pan · covers guideline item **2e**

***Say*** *I'm Pan. Buyers must hear what happened without commerce calling the notification service. The domain event is our Observer. The outbox writer stores an integration event in the same transaction, so it is never lost or announced after a rollback. A relay publishes it every two seconds: to SQS locally, or SNS then SQS on ECS. Delivery is at least once, so the consumer is an Idempotent Receiver: it inserts the event ID with ON CONFLICT DO NOTHING, and a duplicate writes nothing. Dev has no broker; this runs on ECS and in our tests.*

≈ 95 words · ≈ 44 s at 130 wpm · slot 45 s · if short of time: If over time, drop ", so it is never lost or announced after a rollback".

**Cues**

- `[VERIFY: Pan confirms against his draft]` No draft in docs/swe5006/report-drafts/; wording from the code and backend ADR 0007.
- Code: IntegrationOutboxWriter (@TransactionalEventListener BEFORE_COMMIT, lowest precedence, only when blocbuy.events.enabled) → sys_outbox; OutboxRelayJob (fixedDelay 2 s, 2-minute lease) and OutboxRelayService (batch 100, 10 attempts); SysConsumedMessageMapper.xml "ON CONFLICT (event_id, consumer) DO NOTHING"; NotificationEventAppService.handle writes the marker and the notices in one transaction; DLQ after 5 receives (ADR 0007).
- Dev: application-ec2.yml defaults BLOCBUY_EVENTS_ENABLED, relay and consumer to false, and ADR 0007 records the scope decision of 11 Oct (notifications serve SWE5001 only), so dev produces no notices; do not demo them on dev.
  `[VERIFY: the dev host env file leaves BLOCBUY_EVENTS_* off]`
- Tests: OutboxRelayElasticMqIT, NotificationConsumerElasticMqIT, ConsumedMessageRepositoryIT (firstMarkWinsAndRepeatsAreNoopsPerConsumer, concurrentFirstDeliveriesInsertExactlyOneMarker).

**Hand-over** *"Lim, the buyer's own page."*

## Slide 12 UC8 My blocs: the URL owns the view state

16:38:15–16:39:00 · 45 s

**Speaker** Lim · covers guideline item **2e**

***Say***

*Last, My blocs: the buyer's teams and orders on one page. A filter chip and a page number must survive a reload, the Back button and a shared link, and the API has no status filter. So the URL is the single source of view state. A schema validates the search parameters and falls back on bad values. The loader prefetches both lists, and switching a chip filters the cached list without refetching. One adapter function unwraps the response envelope at a single boundary. Liu Jin will now show how we tested all this.*

≈ 94 words · ≈ 43 s at 130 wpm · slot 45 s · if short of time: If over time, drop "One adapter function unwraps the response envelope at a single boundary." (it stays on the slide).

**Cues**

- `[CONFIRM: Lim picks the UC8 design problem]` Proposal from the code: URL as the single source of view state, plus one Adapter at the API boundary. The use-case section of docs/swe5006/report-drafts/lim-jiechao-contribution.md is still "to be written".
- Code: frontend/src/routes/_authenticated/account.tsx (validateSearch: accountSearchSchema, stripSearchParams defaults, loaderDeps page and size only, loader prefetches orders and blocs, redirect past the last page); features/blocs/blocs-search.ts (v.fallback to "active"); my-teams-query.ts (size 100, select unwrapResult); bloc-status.ts. Frontend ADRs 0001 (router owns URL state, Query owns server state), 0003 (generated client), 0009 (unwrapResult as select).
- Tests: frontend/src/routes/account.test.tsx: "switches chips through ?blocs= without refetching", "falls back to Active for an unknown ?blocs= value", "sends a page past the last one to the last page".
- Hand-over: segment 3 (testing) starts 16:39:00 with Liu Jin.

**Hand-over** *"Over to Liu Jin for testing."*

**Watch out**

If a lecturer asks a long design question here, answer the decision in one sentence and park the rest: segment 2 has no slack, and Liu Jin must start testing at 16:39:00.

### Q&A (during the talk / 17:10–17:15)

| Likely question | Answer: decision → reason → evidence or limitation (30–45 s) | Owner |
|---|---|---|
| Why a monolith on dev and services on prod? | Decision: one codebase, two topologies (backend ADR 0008). Reason: SWE5006 grades a complete runnable system, so dev stays the monolith on one EC2 host; SWE5001 needs three services on ECS. The services are thin assemblies of the same modules, and ArchUnit holds the service boundaries in the monolith too. Limitation: commerce still holds four contexts; extracting payment would need an outbox plus a saga. | Eric |
| Is ProductStatus really the State pattern, or an enum switch? | Decision: State, with enum constants as the concrete states. Reason: each constant overrides only the moves it allows and returns the next state; the defaults refuse with 409, and Product delegates without branching on status. Evidence: ProductStatus.java, ProductStatusTest. Contrast: GroupDeal uses a transition matrix (DealStatus.canTransitionTo), which we do not call State. | Liu Jin |
| How do you stop a double refund? | Decision: a durable obligation, then an idempotent call. The obligation is unique per payment and commits with the expiry; the Stripe idempotency key is built from its ID; a lease plus a version fence stops two workers; after 23 hours the worker only searches Stripe, never creates. Evidence: RefundServiceTest.lostResponseRetriesSameDurableKey. Limitation: proven locally with a provider substitute, not on dev (#223). | Yorik |

| | | |
|---|---|---|
| Why a fixed 30 s retry and not exponential backoff? | Decision: a fixed delay in RefundMapper.xml. Reason: the idempotency key makes each retry safe, and the 23-hour cutoff ends automatic retries. Limitation: there is no exponential backoff or jitter, so we call it a fixed delay, not "Retry with Backoff" as issue #463 words it. | Yorik |
| Why a Factory Method with only one creator? | Decision: keep the creator as a seam. Reason: initiate is final, so any creator keeps the validation, and constructor injection (#421) lets a test or configuration supply another creator without editing the service. Limitation: one creator, chosen at build time; we do not claim runtime-pluggable creation. | Eric |
| Where is the analysis model? | Slide 3 is the use-case model. Each member's report section has an analysis class and sequence diagram before the design ones (drafts in docs/swe5006/report-drafts/). Backup B7 maps analysis classes to design classes and ports per member; B1 is the context map and B2 the aggregates. | Eric |
| Is there a database per service? | On dev: one PostgreSQL database, schema "domain", all 29 tables. On ECS: one RDS instance with a database per service, and each runtime role is granted only its own tables (ADR 0008). Every database still applies the full Flyway stream while SWE5006 adds migrations, so foreign tables exist there but stay empty. Full list: B5. | Liu Jin |
| Do notifications work on dev? | No, by decision: the dev host runs no broker and leaves the outbox writer, relay and consumer off (application-ec2.yml; ADR 0007 scope decision, 11 Oct). The path runs on ECS through SNS and SQS, and locally and in CI against ElasticMQ (OutboxRelayElasticMqIT, NotificationConsumerElasticMqIT). | Pan |

# 3 Types of Testing Conducted (3+0 minutes)

| | |
|---|---|
| **Clock** | 16:39:00 – 16:42:00 (3 min talk + 0 min Q&A; talk ends 16:42:00) |
| **Owner** | Liu Jin |
| **Slides** | 13, 14 |

**Lecturer's presentation sequence · Presentation Guidelines · Briefing deliverables (V1.0)**

**3a** — Types of Testing Conducted
**3b** — Well structured source code with unit test cases for classes with more complex behaviours.

**Tip**

Liu Jin has two 90-second slides. At 16:41, while slide 14 is on, Pan (muted) opens the demo pull request; Liu Jin does not wait for it. The hand-over line on slide 14 starts Pan's segment; the share switch to Pan comes later, at 16:46:15.

## Slide 13 Seven kinds of tests, from domain rules to the deployed site

16:39:00–16:40:30 · 90 s

**Speaker** Liu Jin · covers guideline item **3a**

***Say***

*Thanks, Lim. We test at seven levels; the pyramid shows how many of each. At the base, 182 unit test classes cover the domain and application layers, aimed at the classes with the most complex behaviour. RefundServiceTest proves that a lost Stripe response is retried with the same idempotency key, and PaidGroupDealTest checks 24 rules of a paid team. Above them, 49 integration test classes start real PostgreSQL 18, Redis and ElasticMQ in Testcontainers, so unique keys, version checks and the outbox run against the real database, not a mock. Eighteen ArchUnit rules hold the module boundaries, and the contract check regenerates the OpenAPI file from code and fails a breaking change.* **[click]** *On the frontend, 76 Vitest files render components against MSW handlers generated from that same contract. Seven Playwright specs run on every pull request, and again against dev after each deploy. Performance is a scripted load test that we run by hand.*

≈ 154 words · ≈ 71 s at 130 wpm · slot 90 s · if short of time: If over time, drop "and PaidGroupDealTest checks 24 rules of a paid team".

**Cues**

- `[VERIFY: Liu Jin confirms the testing story]` No testing draft in docs/swe5006/report-drafts/; every number below was counted in the repository on 19 Oct at 8bcb38dd.
- Counts (19 Oct): find backend -name "*Test.java" = 183 (182 unit + ArchitectureTest); "*IT.java" = 49 (45 of them start Testcontainers: postgres:18, redis:8.10, softwaremill/elasticmq-native:1.7.1); grep -c @ArchTest ArchitectureTest.java = 18; frontend/src "*.test.ts*" = 76; frontend/e2e = 7 specs, 18 test() declarations. Static @Test/@ParameterizedTest annotations: 1,281 in *Test, 241 in *IT. Per-module split: backup B10.
- Examples: RefundServiceTest.lostResponseRetriesSameDurableKey (15 tests); PaidGroupDealTest (24); ProductStatusTest (5); OrderRepositoryIT.duplicatePaymentIsRejectedAtTheDatabase, saveIfVersionRejectsStaleWriteButAcceptsFreshOne; OutboxRelayElasticMqIT.publishesEventsWithAttributesAndRecoversAfterTransportOutage; StripeRolloutAcceptanceIT (8; real HTTP, scheduler, Flyway and webhook signatures, only OAuth and Stripe network calls substituted).
- Where it runs: ci-backend.yml build-quality (mvnw verify: unit, ArchUnit, ITs) and contract-check (export, git diff --exit-code contracts/, openapi-diff 2.1.7 --fail-on-incompatible unless labelled contract-change-approved); ci-frontend.yml frontend-check (pnpm check incl. Vitest), e2e-preview (PRs), e2e-dev (after deploy-s3-dev). Performance: backend/perf/workload.mjs, docs/perf/sprint3-baseline.md (local, 30 Sep); perf/k6 is the SWE5001 ECS load tool, only its tooling is tested in CI (perf-tooling.yml). Security scans (gitleaks, Trivy) are not counted here: Pan covers them.

**Hand-over** *"So what stops a bad change from merging?"*

## Slide 14 Gates that block a merge, and what tests caught

16:40:30–16:42:00 · 90 s

**Speaker** Liu Jin · covers guideline item **3b**

***Say***

*Tests only matter if they can stop a merge. On dev, six checks are required: the backend build with every test, the Docker image boot, gitleaks, the frontend check, and SonarCloud for both halves; the branch must be up to date first. SonarCloud blocks new code under 80 percent coverage or over 3 percent duplication. Today dev reads 99.2 percent on new backend code and 96.2 on frontend, and our latest full backend run passed 1,931 tests.* **[click]** *Three things tests caught before production. Two red tests showed that a team with no members passed the quote, so a buyer could pay and then be refunded; that fix is on slide 9. ArchUnit failed the build when a new internal controller skipped input validation. And after a deploy, the dev smoke test got a 502: it ran while the single backend host restarted, so our dev deploys are not zero-downtime. Our limits are on the slide; Pan will now show the pipeline that runs all of this.*

≈ 166 words · ≈ 77 s at 130 wpm · slot 90 s · if short of time: If over time, drop the ArchUnit example (it stays on the slide).

**Cues**

- Required checks: gh api repos/PPsyrius/iss-groupbuy-team8/rulesets/22734681 (read 19 Oct): build-quality, SonarCloud Backend, docker, gitleaks, frontend-check, SonarCloud Frontend; strict (up to date); 1 approval, threads resolved, squash only. contract-check, e2e-preview and Trivy run on PRs but are not required.
- Gate and coverage: docs/quality/quality-gates.md, NFR-06 (enforced). SonarCloud API, dev branch, read 19 Oct: backend gate OK, analysis 16 Oct (1f7f4d6a), coverage 99.4% (lines 99.6, branches 99.0), new code 99.2%; frontend gate OK, analysis 19 Oct (d9ff9e9d), coverage 96.1%, new code 96.2%; thresholds new_coverage < 80 and new_duplicated_lines_density > 3 fail. No repository-wide jacoco:check, by decision.
- Totals: docs/agile/sprint-04/log/2026-10-16.md (./mvnw -B verify, 1,931 tests, 0 failures, 1 opt-in skip, local with Docker on 24b4f47); 2026-10-19.md (pnpm check 799 tests; pnpm e2e 17 passed, 5 skipped). Cross-check: CI build-quality run 39132616205 (18 Oct, PR test/auth-surface) summed 1,962 tests, 1 skipped, in 12:36 min.
- Caught: (1) sprint-03/log/2026-10-07.md "two zero/negative-count red tests reproduced the quote defect before the fix" (GroupDealPurchaseSupportImplTest.aTeamWithoutItsPaidLeaderCannotBeQuoted; #422, merged in #424). (2) sprint-04/log/2026-10-13.md: controllers_are_validated required @Validated on InternalUserController (#468). (3) ci-frontend runs 37102432667 (3 Oct) and 37182996807 (4 Oct): e2e-dev smoke got 502 from /api/actuator/health at 06:17:51 and 06:33:03 UTC, inside deploy-existing-ec2 (06:17:23–06:18:03, 06:32:35–06:33:16) of the same commits.
- Limits: test-first for selected changes only (sprint-03/log/2026-10-03.md: rate-limit checks failed 429 vs 200 first; 2026-10-07.md); mutation checks only as hand probes (2026-10-12.md, 2026-10-13.md); load tests by hand; CodeQL job exists but repo variable ENABLE_CODEQL=false (security.yml).

**Hand-over** *"Pan, over to you for the pipeline demo."*

**Watch out**

Recount the test numbers on 25 Oct (open item 13). Never present the 30 Sep workload numbers as a dev or production result: they are local.

## Q&A (during the talk / 17:10–17:15)

| Likely question | Answer: decision → reason → evidence or limitation (30–45 s) | Owner |
|---|---|---|
| How do you test payments without calling Stripe? | Decision: Stripe sits behind gateway ports (Checkout and Refund gateways). Reason: unit tests swap the port; StripeRolloutAcceptanceIT (8 tests) runs the real HTTP endpoints, scheduler, Flyway schema and webhook signature checks, with only OAuth and the Stripe network calls substituted, including the lost-response crash window (local rehearsal, 3 Oct). Limitation: no live Sandbox refund on dev yet (#223, #517). | Liu Jin → Yorik |
| Why Testcontainers rather than H2? | Decision: real PostgreSQL 18, Redis 8.10 and ElasticMQ in containers (45 of 49 IT classes). Reason: our invariants live in the database: UNIQUE keys, ON CONFLICT DO NOTHING, version-guarded UPDATEs, role grants; H2 would accept or reject different SQL. Evidence: OrderRepositoryIT.duplicatePaymentIsRejectedAtTheDatabase, ServiceGrantsIT (5/5 fail with the REVOKE disabled, 12 Oct log). Cost: Docker is needed; the CI build took 12:36 min on 18 Oct. | Liu Jin |
| Do you do TDD? | Not for everything, and we say so. Test-first for selected changes: the rate-limit checks failed (429 expected, 200 actual) before the code (3 Oct log); two red tests reproduced the join-quote defect before the fix (7 Oct log). Elsewhere tests come in the same pull request, and the 80% new-code gate makes them mandatory. | Liu Jin |
| What is your test coverage? | SonarCloud on dev, read 19 Oct: backend 99.4% overall and 99.2% on new code (analysis 16 Oct); frontend 96.1% overall and 96.2% on new code (19 Oct). The gate is at least 80% on new code (NFR-06); there is no repository-wide floor, by decision (quality-gates.md). Limitation: coverage is not assertion quality, and we have no automated mutation testing. | Liu Jin |
| How do you test the frontend against the API? | Decision: contract first. The backend regenerates contracts/openapi.json and fails on drift; openapi-diff fails a breaking change unless labelled. The frontend generates its client and its MSW handlers from that file (frontend ADRs 0003, 0011), and an unhandled request fails a test. Playwright mostly uses shared stubs; only smoke and the @auth/@admin cases reach the real backend. Limitation: stubs prove the frontend half only. | Liu Jin → Lim |
| Do you do performance testing? | Yes, by hand, not in CI. Local baseline 30 Sep (docs/perf/sprint3-baseline.md): 100 virtual users per scenario on real HTTP, PostgreSQL and Redis; p95 24 ms browse, 57 ms JOIN on one hot team, 18 ms My Orders, zero errors; 30 team expiries committed within 60 s. Local only, not a deployed result. k6 drives the SWE5001 ECS load test; the system demo shows it. | Eric |

# 4 CI/CD Pipeline Demo (11 minutes)

| | |
|---|---|
| **Clock** | 16:42:00 – 16:53:00 (slides 15–20 until 16:46:15, then the live pipeline demo until 16:53:00) |
| **Owner** | Panpakorn Siripanich (Pan); Lim approves as the frontend code owner |
| **Slides** | 15, 16, 17, 18, 19, 20 |

**Lecturer's presentation sequence · Presentation Guidelines · Briefing deliverables (V1.0)**

**4a** — Describe the strategy to manage the source code and artifacts.
**4b** — Describe the DevSecOps pipeline and procedures.
**4c** — Highlight any technical findings and issues encountered.
**4d** — Design specification of the DevOps pipelines along with the use of appropriate tools with minimal security considerations.
**4e** — Backup: Record a video in advance in case of any technical issues during the live demo.

**Tip**

Slides 15–20 take 255 s. Pan opens the pull request at 16:41, during slide 14, so its checks (median 4.8 min) are green when the live part starts at 16:46:15. The live part shows one real frontend change end to end and the backend deploy as a recorded run: a backend change could not finish its 16-minute PR checks inside the segment.

## Slide 15 Source code and artefacts: one trunk, images by digest

16:42:00–16:42:45 · 45 s

**Speaker** Pan · covers guideline item **4a**

***Say*** *Thanks, Liu Jin. One trunk: dev. Every change is a short feature branch and a pull request with a Conventional Commit title, squash-merged, so history stays linear. The ruleset needs one approval from a code owner: pipelines route to me, backend to Liu Jin, frontend to Lim. Threads must be resolved and six checks pass on an up-to-date branch. Dependabot proposes updates daily. Artefacts: the backend image sits in ECR with immutable tags and is deployed by digest, never by tag; the frontend is one static build with no per-environment settings; one OpenAPI file generates the client.*

≈ 97 words · ≈ 45 s at 130 wpm · slot 45 s · if short of time: If over time, drop "Dependabot proposes updates daily." (it stays on the slide).

**Cues**

- `[VERIFY: Pan confirms the pipeline story]` Pan has no report draft; every fact on slides 15–20 comes from the repository, gh and AWS read-only calls of 19 Oct.
- Ruleset: gh api repos/PPsyrius/iss-groupbuy-team8/rulesets/22734681 (read 19 Oct) "dev-require-review": squash only, 1 approval, require_code_owner_review, dismiss_stale_reviews_on_push, required_review_thread_resolution, 6 strict required checks (build-quality, SonarCloud Backend, docker, gitleaks, frontend-check, SonarCloud Frontend), non_fast_forward, deletion; no bypass actors listed. CODEOWNERS: /.github/ @PPsyrius, /backend/ @l0sgAi, /frontend/ @limjiechao. Conventional Commits: CONTRIBUTING.md §Commits; PR template .github/PULL_REQUEST_TEMPLATE.md (evidence checklist).
- Dependabot: .github/dependabot.yml (maven, github-actions, docker, npm; daily; 7-day cooldown; Spring, Java, React and TypeScript majors ignored); 70 Dependabot PRs merged (slide 4). Tags: git tag = v0.1.0-sprint1/2/3; a v* tag runs the full ci-backend gate chain.
- Artefacts: ECR dev/blocbuy IMMUTABLE and scanOnPush (AWS CLI, 19 Oct); deployment/backend/ecr-lifecycle-policy.json keeps the newest 30; ci-backend.yml publish-ecr-dev pushes git-<sha>-run-<n> and checks the digest against ECR; deploy-ec2 action rejects a URI not pinned by sha256. Frontend: ADR 0004 (no VITE_*, CI grep), ci-frontend.yml deploy-s3-dev. Contract: contracts/openapi.json, contract-check; frontend ADR 0003 (generated client, not committed).
- SWE5001 prod branch: ruleset 24551375 "prod-release" (promotion PR only, merge commit, 3 required checks: promotion-source, service-images, infra-plan). Same project, other topology; say "SWE5001" if asked.

**Hand-over** *"Here is how a change travels."*

## Slide 16 One pipeline from pull request to dev

16:42:45–16:43:45 · 60 s

**Speaker** Pan · covers guideline item **4b**

***Say***

*On a pull request, the backend builds and runs every test, checks the API contract, then builds the image, boots it, checks its health endpoint and scans it with Trivy. SonarCloud blocks new code under 80 percent coverage. The frontend runs pnpm check, SonarCloud, and Playwright on its own build, and every pull request gets a secret scan. The blue boxes are the six required checks: about sixteen minutes for backend, five for frontend. After the squash merge, delivery rebuilds from the merged commit. The backend image goes to ECR by digest and reaches the host through SSM, with a health check and automatic rollback. The frontend syncs to S3 and is smoke-tested through CloudFront. Either is live in about three minutes.*

≈ 122 words · ≈ 56 s at 130 wpm · slot 60 s · if short of time: If over time, drop "SonarCloud blocks new code under 80 percent coverage." (Liu Jin covered it on slide 14).

**Cues**

- Diagram: src/diagrams/cicd-pipeline.svg, checked against .github/workflows/ci-backend.yml, ci-frontend.yml, security.yml and .github/actions/deploy-ec2/action.yml on 19 Oct.
- Timings, gh (19 Oct), medians of successful runs: backend PR start → docker end 964 s (18 runs, 16–19 Oct; build-quality 785 s, docker 170 s, contract-check 12 s); frontend PR start → e2e-preview end 289 s (19 runs; frontend-check 206 s, e2e-preview 69 s). Push to dev: backend start → deploy-existing-ec2 end 186 s (88 runs, 14 Sep–18 Oct; package-dev 46 s, publish-ecr-dev 73 s, deploy 40 s); frontend start → e2e-dev end 159 s (94 runs, 30 Sep–19 Oct; package 35 s, S3 21 s, e2e-dev 82 s). Example runs: backend deploy 39133506399 (18 Oct) 3 m 23 s = 55 s + 1 m 28 s + 43 s; frontend 39192166301 (19 Oct) merge → e2e-dev done 2 m 40 s, whole run 4 m 38 s because the post-merge Sonar scan runs alongside; backend PR 39132639003 16 m 29 s.
- Success, last 100 push runs: backend deploy job 88 of 97 (all 9 failures 14–17 Sep: SSM SendCommand InvalidInstanceId "Instances not in a valid state"; none since); frontend deploy-s3-dev 97 of 97, e2e-dev 94 of 97 (3–4 Oct, the 502s of slide 14). Full table: B12.
- Honest detail: the deployed jar/image and dist/ are rebuilt from the merged commit, not promoted from the PR run; Docker smoke and Trivy are not re-run after merge (docs/ci/backend-pipeline.md "Post-merge delivery boundary"). Strict up-to-date checks mean the merged tree is the tested tree; ECR scans the pushed image. Frontend ADR 0004 and the EC2 operator guide still describe promotion of the tested artefact: open item 18.
- Merge queue: merge_group triggers exist but are dormant (private personal repository, ci-frontend.yml comment). Deploy environments backend-dev and frontend-dev allow only dev and have no required reviewer, so the merge is the release decision.

**Hand-over** *"Security runs through every one of those boxes."*

## Slide 17 Security in the pipeline, no stored keys

16:43:45–16:44:30 · 45 s

**Speaker** Pan · covers guideline item **4d**

***Say*** *Security sits in each stage. Gitleaks is required on every pull request; SonarCloud and SpotBugs analyse the code; Trivy scans dependencies, configuration and the image. Every third-party action is pinned to a commit hash. There are no AWS keys in GitHub: three OIDC roles, trusted only from dev, each scoped to one job: push to one repository, run one command document on one instance, or sync one bucket. The container runs read-only with every capability dropped, and the account sits under organisation guardrails. Our gaps are on the bottom line.*

≈ 90 words · ≈ 42 s at 130 wpm · slot 45 s · if short of time: If over time, drop "Every third-party action is pinned to a commit hash." (it stays on the slide).

**Cues**

- Controls: security.yml (gitleaks on every PR and dev push; trivy-deps, trivy-config, workflow-lint, deployment-bundle path-filtered on PRs, all on v* tags and weekly Monday 02:17 UTC); ci-backend.yml build-quality (SpotBugs, Checkstyle, ArchUnit), docker (Trivy image HIGH/CRITICAL, ignore-unfixed, exit-code 1). 28 external actions, every one pinned to a 40-character SHA (grep of .github/workflows and .github/actions, 19 Oct).
- OIDC roles verified on AWS 19 Oct (orchestrator): BlocBuyGitHubEcrPush (trust: dev branch), BlocBuyGitHubBackendDeploy (trust: backend-dev environment; ssm:SendCommand on AWS-RunShellScript and one instance per docs/deploy/backend-ec2-pipeline-operator-guide.md §6 and deployment/backend/iam/github-ssm-deploy-policy.json.example), BlocBuyGitHubFrontendDeployDev (trust: frontend-dev environment; one bucket and one distribution). Both environments allow only the dev branch (gh api, 19 Oct).
- Runtime: deployment/backend/compose.yml read_only, cap_drop ALL, no-new-privileges, /tmp tmpfs, host network; secrets only in /etc/blocbuy/backend.env root:root 0600 (operator guide §7). Account: docs/deploy/aws-account-governance/README.md (Org-Baseline and Region-Pin on Root, Sandbox-Guardrails on the Sandbox OU; Identity Center, 1-hour sessions).
- Gaps: ENABLE_CODEQL=false (gh variable list, 19 Oct) and the code-scanning API answers "Code scanning is not enabled for this repository"; NFR-37 / #227 open (docs/deploy/dev-cloudfront-deployment.md §13); dev EBS unencrypted (AWS, 19 Oct); Gitleaks scans event-selected commits and a 14 Oct full-history audit left 24 old matches undecided (docs/quality/quality-gates.md). Two IAM users for SSM port forwarding, one active key (governance README): CI has no keys, the account does. Full list: B14.

**Hand-over** *"And this is where it all lands."*

## Slide 18 Where it runs: dev on EC2, prod on ECS

16:44:30–16:45:15 · 45 s

**Speaker** Pan · covers guideline item **4b**

***Say*** *This is where dev runs, all in Singapore. The browser only talks to CloudFront. Static files come from a private S3 bucket through origin access control. API calls go to a load balancer that accepts only CloudFront's address range and answers 403 without our origin-secret header. Behind it, one EC2 host runs the backend container from Docker Compose, next to PostgreSQL and Redis on localhost. Logs go to CloudWatch for fourteen days. On the right is SWE5001's production: the same code as three services on ECS Fargate, rebuilt on every release.*

≈ 91 words · ≈ 42 s at 130 wpm · slot 45 s · if short of time: If over time, drop "for fourteen days".

**Cues**

- Diagram: src/diagrams/dev-deployment.svg. AWS facts read-only on 19 Oct (orchestrator; aws --profile blocbuy --region ap-southeast-1 describe/get/list): CloudFront E3U5W7UWDL8LRV d1b3np88z657cp.cloudfront.net, HTTP/2, no WAF, origins S3 blocbuy-dev-frontend-402178232802 (/*) and ALB blocbuy-dev-backend (/api/*); ALB internet-facing, 2 AZs, HTTP:80, default fixed 403, one forward rule on an http-header condition; SG ingress 80 only from com.amazonaws.global.cloudfront.origin-facing; target group HTTP 8589, health /api/actuator/health; EC2 i-0852b6c3475b19f05 "BlocBuy-dev" t3.large x86_64, ap-southeast-1b, Ubuntu 24.04, SSM online, 40 GB gp3; ECR dev/blocbuy; log group /blocbuy/backend/dev, 14 days; no CloudWatch alarm for dev (all six are SWE5001's). Re-checked read-only on 19 Oct after sign-in: S3 origin behind OAC E2WEBPIBHXOW90; ALB origin http-only (NFR-37); viewer redirect-to-https with a response headers policy; EC2 IMDSv2 required, public IP, host security group sg-04358da6b0bc71343 allows 8589 only from the ALB group and SSH 22 from 0.0.0.0/0 (B14); OIDC trust pinned to environment:backend-dev, environment:frontend-dev and refs/heads/dev; the deploy role may only SendCommand AWS-RunShellScript to i-0852b6c3475b19f05; volume unencrypted; 0 dev alarms.
- Repo sources: docs/deploy/dev-cloudfront-deployment.md (OAC blocbuy-dev-frontend-oac, X-Origin-Secret rule, security headers policy, NFR-37); docs/deploy/backend-ec2-pipeline-operator-guide.md §1 (PostgreSQL and Redis on 127.0.0.1, pre-existing, not pipeline-managed; the Compose file holds only the backend, enforced by security.yml deployment-bundle); docs/deploy/backend-log-observability.md (PostgreSQL 18, Redis 8 on the host).
- Not shown: a second CloudFront dcsjrkmummxi7.cloudfront.net → S3 blocbuy-media-dev-402178232802 with a WAF web ACL exists in the account, but no document or code in the repository names it (grep, 19 Oct): open item 17. Stripe and Google are external callees of the backend (B13 shows them).
- Prod box: SWE5001 topology (shared physical-deployment diagram, backup B13; backend ADR 0008).

**Hand-over** *"Here is what building this taught us."*

## Slide 19 Technical findings and issues

16:45:15–16:46:00 · 45 s

**Speaker** Pan · covers guideline item **4c**

**Say** *Four findings. First, a green pipeline is not a working feature: on dev, Stripe Checkout opens and returns, but the webhook never confirms the payment. That's issue 517, still open. Second, every backend deploy restarts the only host, so a smoke test once got a 502; we accept that on dev. Third, the organisation guardrails also block us as operators, so IAM changes go through the CLI. Fourth, from SWE5001: tests with mocks passed, but the real cloud failed, on a missing permission and a read-only temp folder.*

≈ 88 words · ≈ 41 s at 130 wpm · slot 45 s · if short of time: If over time, drop the third finding (guardrails) from speech; it stays on the slide.

**Cues**

- (1) #517 open (16 Oct: payment 01a14345-… stays waiting; likely causes in the issue: Sandbox endpoint URL, signing secret or subscribed events); #223 open (Checkout, scheduling and expiry refunds on dev). Owner Yorik; go/no-go 23 Oct (open item 1).
- (2) ci-frontend runs 37102432667 (3 Oct) and 37182996807 (4 Oct): e2e-dev got 502 from /api/actuator/health inside deploy-existing-ec2 of the same commit (slide 14). Decision: accept for dev (single host, container replaced in place, compose.yml stop_grace_period 30 s); not re-tested since.
- (3) docs/agile/sprint-02/log/2026-09-23.md "SCPs block bucket-list and Access Analyzer views but not scoped deployment"; docs/deploy/backend-log-observability.md troubleshooting (Access Analyzer denied → apply IAM with aws iam create-policy-version); consolidated to three SCPs on 16 Oct (#530, governance README).
- (4) SWE5001: PR #527 (16 Oct): Terraform tests on mocks passed, the real Up run 38888232111 failed because the apply role lacked application-autoscaling:TagResource; docs/agile/sprint-03/log/2026-10-06.md: read-only root on Fargate left /tmp owned by root, Tomcat could not start; fix VOLUME ["/tmp"] in both Dockerfiles. The same image worked on the dev host because Compose mounts a 1777 tmpfs.
- Also true, in B12: the first nine dev deploys (14–17 Sep) failed when SSM refused the instance ("not in a valid state"); none of the 88 since failed.

**Hand-over** *"Now let me show you one change going through."*

## Slide 20 Live: one change from pull request to dev

16:46:00–16:46:15 · 15 s

**Speaker** Pan · covers guideline item **4b**

**Say** *Now live: one small frontend change. Checks take about five minutes; after approval and a squash merge, it's on dev in under three. The backend run and a fallback are recorded.*

≈ 31 words · ≈ 14 s at 130 wpm · slot 15 s

**Cues**

- Opens the live pipeline demo (about 405 s, run book in the script). `[CONFIRM: the demo change, its author and the approving code owner (frontend → Lim)]` Open the PR before 16:42 so its checks finish during slides 15–19.
- Numbers: frontend PR checks median 289 s (19 runs, 16–19 Oct); merge → e2e-dev green median 159 s (94 runs); backend merge → deployed median 186 s (88 runs). Recorded backend run: 39133506399 (18 Oct).
- Item 4e: `[FILL: fallback recording link and date (due 25 Oct, open item 19)]`

**Hand-over** (demo) "Here is the pull request."

## Live pipeline demo (16:46:15 – 16:53:00, ≈ 405 s)

| Step | Shows | Items |
|---|---|---|
| P1 | The pull request and its required checks | Items **4b** (procedure), **4a** (branch, PR, Conventional Commit) |
| P2 | Code-owner approval | Items **4a**, **4d** (review gate) |
| P3 | Squash merge | Item **4a** (linear history) |
| P4 | Delivery runs; the backend deploy, recorded | Items **4b**, **4d** (digest, SSM, health check) |
| P5 | Rollback, by design only | Item **4d** (recovery procedure, honest limit) |
| P6 | The change is live on dev | Item **4b** (push to live), maintainability |
| P7 | Wrap up and hand over | Items **4a–4d** summary |

### Preconditions (checked at 15:30 by Pan)

- Demo branch `feature/frontend-swe5006-live-demo` pushed and up to date with `origin/dev` (A.5); PR body file ready; no PR opened yet. `[CONFIRM: the copy change, its page and the approving code owner (open item 15)]`
- Merge freeze announced in the team chat: nobody merges into `dev` from 15:00 to 17:15 except this PR. A moved `dev` makes the branch out of date and re-runs the checks (strict required checks).
- Latest `ci-frontend` and `ci-backend` runs on `dev` green (A.1); deploy switches `ENABLE_S3_DEPLOY` and `ENABLE_EC2_DEPLOY` unchanged.
- Lim signed in to GitHub on his own machine with the PR list open; he is not sharing at 16:47.
- Fallback: the 25 Oct recording, cut into clips P1–P7, open in a player on Pan's machine and tested with Zoom "share sound" off. `[FILL: recording link (open item 19)]`

### Screen layout (Pan's browser window, shared from 16:46:15)

| Tab | Content | Prepared state |
|---|---|---|
| 1 | GitHub: the demo pull request | Opened at 16:41; Files changed tab ready |
| 2 | GitHub Actions: `ci-frontend` filtered to branch `dev` | Refresh after the merge to see the new run |
| 3 | GitHub Actions run 39133506399 (backend deploy, 18 Oct) | Run graph open; deploy job one click away |
| 4 | `.github/workflows/rollback-backend.yml` on `dev` | File view, not the Run workflow dialog |
| 5 | Dev site `https://d1b3np88z657cp.cloudfront.net` | On the page the change touches; DevTools closed |
| Player | Clips P1–P7 (25 Oct) | Paused at the start of clip P1 |

## Step by step

### Step P1 The pull request and its required checks — 16:46:15 – 16:47:15

**Proves** Items **4b** (procedure), **4a** (branch, PR, Conventional Commit) · Pan

**Operate**
1. Tab 1: the demo PR opened at 16:41 → **Files changed**: the one-line frontend change.
2. Switch to **Checks**: frontend-check, SonarCloud Frontend, gitleaks, e2e-preview; the backend jobs report success after skipping (no backend path changed).
3. Point at the merge box: review required from a code owner.

**Say** *This is the pull request I opened at 16:41: one line in the frontend, on a short branch with a Conventional Commit title. These are the required checks from slide 16: the frontend check with all unit tests, SonarCloud, the secret scan, and Playwright against this build. The backend jobs see no backend change and pass straight through. The merge button stays locked until a code owner approves.*

**Expect** Checks green or finishing (median 4.8 min from 16:41, so ≈ 16:46); merge box asks for a code-owner review.

**Fallback** Checks still running: keep talking through them and move on at 16:47:15 regardless. A check red: say what it caught in one sentence, then `CLIP P2` (the 25 Oct recording from approval onwards); leave the PR open and never merge it on air.

### Step P2 Code-owner approval — 16:47:15 – 16:48:00

**Proves** Items **4a**, **4d** (review gate) · Pan; Lim approves

**Operate**
1. Pan posts `APPROVE`. Lim (not sharing) approves from his own GitHub session: **Review changes → Approve**, or `gh pr review <n> --approve`.
2. Tab 1: refresh; the approval from @limjiechao appears; the merge box turns green once the checks are green.

**Say** *Frontend changes belong to Lim in CODEOWNERS, so his approval is the one required; mine would not count. If anyone pushes again, the approval is dismissed and the checks run again.*

**Expect** One approving review by a code owner; all checks passed; squash merge enabled.

**Fallback** Lim cannot approve within 30 s: nobody else is a frontend code owner, so do not ask anyone else. `CLIP P2`.

### Step P3 Squash merge — 16:48:00 – 16:48:30

**Proves** Item **4a** (linear history) · Pan

**Operate**
1. Tab 1: **Squash and merge**, keep the Conventional Commit title, confirm. Only this PR.
2. Say the merge time aloud (≈ 16:48:15).

**Say** *Squash merge: one commit on dev carrying the pull request number, so history stays linear. That push starts delivery.*

**Expect** PR merged; leave the branch-delete button alone.

**Fallback** "Base branch was modified" (dev moved): do not update the branch on air; it would re-run five minutes of checks. `CLIP P3`; merge after 17:15.

### Step P4 Delivery runs; the backend deploy, recorded

16:48:30 – 16:50:00

**Proves** Items **4b**, **4d** (digest, SSM, health check) · Pan

**Operate**

1. Tab 2: Actions → `ci-frontend` → the new run on `dev`: package-dev → deploy-s3-dev → e2e-dev.
2. Tab 3: the recorded backend run 39133506399 (18 Oct, 3 m 23 s): package-dev 55 s → publish-ecr-dev 1 m 28 s → deploy-existing-ec2 43 s. Open the deploy job: the image is pinned by digest.

***Say*** *Delivery rebuilds from the merged commit. The frontend goes to S3, index.html is invalidated in CloudFront, then Playwright smoke-tests the real site. While it runs, here is a backend delivery from Sunday: the image is pushed to ECR, deployed by digest through SSM to the one EC2 host, health-checked, and rolled back automatically if the check fails. Three minutes twenty-three, start to finish.*

**Expect** Frontend run in progress (median 2.7 min from push to e2e-dev green); recorded backend run green.

**Fallback** Run not started within 60 s: refresh once; then `CLIP P4`.

### Step P5 Rollback, by design only

16:50:00 – 16:50:30

**Proves** Item **4d** (recovery procedure, honest limit) · Pan

**Operate**

1. Tab 4: `.github/workflows/rollback-backend.yml` on dev: its inputs (target, reason, confirmation) and the shared, non-cancellable concurrency group. Never open the Run workflow dialog.

***Say*** *For a manual rollback there is this workflow: it needs a reason and a confirmation and shares the deploy lock. To be honest, it has never run, and no automatic rollback has fired yet. It is a design we have not exercised.*

**Expect** Workflow file visible; its Actions page lists no runs.

**Fallback** Skip if behind (saves 30 s): slide 19 and B14 already say it.

### Step P6 The change is live on dev

16:50:30 – 16:52:15

**Proves** Item **4b** (push to live), maintainability · Pan

**Operate**

1. Tab 2: e2e-dev green (≈ 16:51).
2. Tab 5: `https://d1b3np88z657cp.cloudfront.net` on the changed page → reload → the new text. `[CONFIRM: the page and the copy change (open item 15)]`
3. Read the elapsed time off the run: merge → e2e-dev green.

***Say*** *Smoke test green, and here is the change on the dev site, about three minutes after the merge. The same pipeline put the version you are about to see in the system demo on this host.*

**Expect** New text visible after a reload (index.html is served no-cache and invalidated); run green.

**Fallback** e2e-dev red: show the failed step honestly, it is the gate doing its job, then `CLIP P6`. Text not visible yet: one more reload, then `CLIP P6`.

### Step P7 Wrap up and hand over 16:52:15 – 16:53:00

**Proves** Items **4a–4d** summary · Pan

**Operate**
1. Stay on tab 5 (the dev site) while speaking.
2. At 16:53:00: stop sharing; Eric starts sharing (SHARE).

***Say*** *So one change went from pull request to dev in about ten minutes, with six required checks, a code-owner review and an automatic smoke test, and no AWS keys stored in GitHub. Eric will now show what that system does for a buyer.*

**Expect** Hand-over on time.

**Fallback** If behind, say only the last sentence.

**Never live**

Clicking Merge on anything other than the demo PR; “Update branch” on the demo PR; re-running a failed job; running `rollback-backend.yml` or any `workflow_dispatch`; opening Settings → Secrets and variables or the ruleset editor; pushing to `dev`; any AWS console or SSM session on the shared screen. If the PR is not merged by 16:50, leave it open and merge after 17:15, so a deploy never lands during the system demo.

**Tip**

Narrate results, not clicks: “checks green in four minutes fifty”, not “now I click the checks tab”. No troubleshooting on air.

## Q&A (during the talk / 17:10–17:15)

| Likely question | Answer: decision → reason → evidence or limitation (30–45 s) | Owner |
|---|---|---|
| Why EC2 and not ECS for dev? | Decision: dev reuses one existing EC2 host with the monolith in Docker Compose (operator guide; backend ADR 0008). Reason: SWE5006 needs one complete, always-on system at low cost, and the monolith fits one t3.large; ECS is the SWE5001 topology, rebuilt per release. Limitation: one host, so a deploy restarts the API (502 on 3–4 Oct), and PostgreSQL and Redis share the host outside the pipeline. | Pan |
| How do you roll back? | Automatic: deploy.sh health-checks the new container and restarts the previous digest if it fails; the workflow also checks public health and rolls back. Manual: rollback-backend.yml (reason, confirmation, same non-cancellable concurrency group). Limitation: rollback-backend.yml has never run (gh, 19 Oct) and no automatic rollback has fired (every red deploy failed earlier, at SSM). Flyway is never reversed, so migrations stay backward-compatible. Frontend: redeploy the previous commit (sync uses --delete). | Pan |
| How are secrets handled? | CI holds no AWS keys: OIDC gives short-lived roles per job. The repository has three Actions secrets, none for AWS or the app: SONAR_TOKEN, TYPESAFE_API_KEY and BOARD_SNAPSHOT_TOKEN (gh secret list, 19 Oct). Dev application secrets live in /etc/blocbuy/backend.env (root, 0600) on the host, never in GitHub or an SSM command; SWE5001 prod uses Parameter Store. Gitleaks gates every PR. Limitations: the host file is maintained by hand, and the account still has two IAM users for SSM port forwarding, one with an active key (governance README). | Pan |
| Why is CodeQL off? | The job exists in security.yml behind ENABLE_CODEQL, which is false: code scanning is not enabled for this private repository (GitHub API, 19 Oct), and the operator guide says to require CodeQL only once it is. SAST is covered by SonarCloud on every PR plus SpotBugs in the build. We list it as a gap (B14). | Pan |
| How long from merge to live? | Medians from gh on 19 Oct: backend about 3.1 minutes from the push run starting to the deploy job ending (88 runs); frontend about 2.7 minutes including the e2e-dev smoke (94 runs). Before that, PR checks take about 16 minutes for backend and 5 for frontend. Per-workflow table: B12. | Pan |

| | | |
|---|---|---|
| What stops a bad deploy? | Before merge: six required checks on an up-to-date branch and a code-owner approval. After merge: the image URI must be an ECR digest in the expected repository, the host health check with automatic rollback, the public health check, and the e2e-dev smoke through CloudFront. ENABLE_EC2_DEPLOY and ENABLE_S3_DEPLOY freeze delivery. Limitation: the delivered image is rebuilt after merge and not rescanned by Trivy (ECR scans on push), and the deploy environments have no required reviewer. | Pan |
| How do you manage dependencies? | Dependabot checks Maven, npm, Docker and GitHub Actions daily with a 7-day cooldown and groups (Spring, test, TanStack, HeroUI); majors of Spring, Java, React and TypeScript are ignored on purpose. Every update passes the same required checks; 70 Dependabot PRs merged. Trivy scans dependencies on change and weekly. Actions are pinned to commit SHAs, so Dependabot bumps the SHA. | Pan → Lim |

# 5 System Demo (14+5 minutes)

| | |
|---|---|
| **Clock** | 16:53:00 – 17:07:00 talk and demo · 17:07–17:10 buffer for questions asked during the talk · 17:10–17:15 Q&A |
| **Owner** | Wang Ziang (Eric) and Yu Qianshuo (Yorik) (drive); Lim closes and chairs the Q&A |
| **Slides** | 21, 22; backups B4, B15 and B16 may be used on air |

**Lecturer's presentation sequence · Presentation Guidelines · Briefing deliverables (V1.0)**

**5a** — Demonstrate some of the significant use cases of the solution.

**5b** — Demonstrate how the solution deals with some of the important quality attributes e.g. scalability (such as demonstrating auto-scaling), performance (show using some scripted performance testing), maintainability (show/run the DevOps pipeline).

**5c** — Backup: Record a video in advance in case of any technical issues during the live demo.

## How the demo meets 5a and 5b

| Step | Shows | Quality attribute · item | Environment |
|---|---|---|---|
| 1 | Start a bloc and share the invite | Functionality, security · **5a** (UC2, UC3, UC7) | live on dev |
| 2 | A friend joins through the invite and pays | Consistency · **5a** (UC4, UC7) | live on dev |
| 3 | My blocs and my orders | Functionality · **5a** (UC8) | live on dev |
| 4 | The 16:05 bloc expires and refunds | Reliability, consistency · **5a** (UC5, UC6), **5b** | live on dev |
| 5 | One seat, one refund: consistency under failure | Consistency, reliability · **5b** | dev outcome; races proven by tests |
| 6 | Scalability and performance, labelled | Scalability, performance · **5b** | SWE5001 prod (ECS) and local, labelled |
| 7 | Maintainability: the pipeline you just saw | Maintainability · **5b** | live on dev (segment 4) |

Use cases (5a) are steps 1–4 on the deployed dev site. The quality attributes named in 5b are covered as the guideline asks: **scalability** by SWE5001 auto-scaling on ECS (step 6, labelled; dev has none by design), **performance** by scripted load tests (step 6: k6 on ECS, the Node workload locally), **maintainability** by the pipeline run live in segment 4 (step 7). Consistency and reliability, central to BlocBuy's money flow, are steps 2, 4 and 5 (slides 9, 10). Item 5c is the 25 Oct recording, one clip per step.

## Principle: start the slow part before the talk

A bloc cannot expire faster than its team lifetime, and the minimum lifetime is the 2,400-second quote window (`ProductTeamTerms.MIN_LIFETIME_SECONDS`), so `US34-04` lives 40 minutes. The expiry sweep runs every minute and the refund worker every 30 s (one obligation per run), so the refund shows ≈ 1–2 minutes after the deadline. Started at **16:05**, the bloc expires ≈ 16:45 and its refund is visible well before step 4 at 16:58:50. Deadlines are never edited on shared dev (`backend/docs/payment/stripe-rollout-runbook.md`): the timer, not a manual call, is the evidence.

## Decision: live or recorded payment steps (go/no-go, Fri 23 Oct, 18:00)

| Condition (Yorik decides with Lim; posted in #464) | GO: live on dev | NO-GO: recorded or local |
|---|---|---|
| #517 fixed: a new START on dev reaches paid within the receipt's one-minute refresh, and the bloc shows under My blocs → Active | Steps 1–3 live; fallback = 25 Oct clips | Steps 1–2 from local evidence (B15, 20 Sep), said as "verified locally, not yet on dev"; step 3 live on existing data |
| #223 done: Checkout and scheduling on; a `US34-04` bloc started on 24 Oct expired within one sweep and its refund reached succeeded on the receipt | Step 4 live with the 16:05 bloc | Step 4 from B15 (local); slide 21 footer changes (open item 23) |
| Full dry run on dev on Sat 24 Oct passes in order, timed | Rehearse at the real clock times on 25 Oct | Only the NO-GO path is rehearsed |
| Late fix (24–25 Oct) | GO only if the dry run and the recordings are done by Sun 25 Oct 12:00 | After 12:00 on 25 Oct the NO-GO stands; no switch is changed on dev on 26 Oct |

**Watch out**

No refund has ever been proven on dev (19 Oct): every dev payment and refund in this run book is conditional on the dry run. A GO for steps 1–3 with a NO-GO for step 4 is allowed; then only step 4 uses B15. `[CONFIRM: go/no-go result of 23 Oct and the slide 21 footer (open items 1, 24)]`

## Demo preconditions (checked at 15:30 by Eric and Yorik)

- Health `UP` (A.1); latest `ci-backend` and `ci-frontend` runs on `dev` green; merge freeze in force.
- Checkout and scheduling enabled on dev (#223) `[VERIFY: how to read the flags on the day without SSM: the capabilities endpoint (A.4) or the operator's day-log entry]`; listings converged: `scripts/seed-dev-demo-listings.sh --check` passes for all four `US34-*` listings (A.2, Yorik, admin token).
- Two Google buyer accounts signed in on Eric's machine in two Chrome profiles, "Leader" (account A) and "Joiner" (account B), and the same two on Yorik's machine as standby (sessions are concurrent and last 24 h). Neither is a member of an open `US34-02` bloc. `[CONFIRM: which two accounts and the display names shown on screen (open item 22)]`
- The 16:05 `US34-04` bloc open at 1 of 2 (`EXPIRY OK` at 16:10); its receipt bookmarked in the Leader window, tab 4. Nobody else has its invite link.
- Fallback clips 1–8 from the 25 Oct recording on Eric's machine, tested with Zoom "share sound" off; a copy on Yorik's. `[FILL: clip folder link (open item 20)]`
- Do Not Disturb on; one monitor shared; browser zoom 125%; no password manager, email or bookmarks bar visible. This run book on a second device, not on the shared screen.

## Screen layout (Eric shares one screen)

| Window · tab | Content | Prepared state |
|---|---|---|
| Leader · 1 | Deck `swe5006-presentation.html#21` | Slide 21; backups one key away (B); slide 22 at the end |
| Leader · 2 | Listing `US34-02` on `https://d1b3np88z657cp.cloudfront.net` | Signed in as account A; product page loaded |
| Leader · 3 | `/account` (account A) | Loaded; the Refunded chip shows the 16:05 bloc from ≈ 16:46 |
| Leader · 4 | Receipt `/payments/{paymentId}` of the 16:05 payment | Bookmarked at 16:10; reload before step 4 |

| | | |
|---|---|---|
| Leader · 5 | Pan's merged PR and the changed dev page | Opened after 16:52 |
| Joiner · 1 | Empty tab for the invite link | Signed in as account B |
| Joiner · 2 | `/account` (account B) | Loaded |
| Player | Clips 1–8 (25 Oct) | Paused at the start of clip 1 |

## Opening (slide 21)

**Slide 21 Live demo: one bloc from invite to refund** 16:53:00–16:53:20 · 20 s

**Speaker** Eric · covers guideline item **5a**

**Say** *Next, one bloc's life on our dev site: a buyer starts it and shares an invite, a friend joins and pays in Stripe Sandbox, both see it under My blocs, and a bloc we started at 16:05 expires and refunds. Then the quality attributes.*

≈ 44 words · ≈ 20 s at 130 wpm · slot 20 s

**Cues**

- 16:53:00: Eric is sharing (switch from Pan at the hand-over); his window shows swe5006-presentation.html#21. Follow the demo run book in the script step by step; Yorik follows on the same page and posts cue words in the backchannel.
- Storyline from #464: invite → START/JOIN with Stripe Sandbox → /account orders and blocs → expiry refund. Steps 1–4 are UC2–UC8 on dev; step 5 shows 5b with each environment named (dev, SWE5001 prod on ECS, or local).
- `[CONFIRM: footer: keep "Live on dev; recorded steps are marked" only after the 23 Oct go/no-go (#223, #517); with a NO-GO, say "the payment steps are recorded" here]`

**Hand-over** *"First, our leader starts a bloc."*

## Step by step

**Step 1 Start a bloc and share the invite** 16:53:20 – 16:55:20

**Proves** Functionality, security · **5a** (UC2, UC3, UC7) · Eric operates and narrates

**Operate**

1. Window "Leader" (account A, signed in with Google), tab 2: listing `US34-02` (team of 3, 24 h; 39.00, team price 29.00).
2. **Start a bloc** → the quote sheet shows the team price → **Continue to payment**.
3. Stripe Checkout with its Sandbox banner: test card 4242 4242 4242 4242, any future expiry, any CVC → Pay.
4. Back on BlocBuy: the receipt turns from "Waiting for payment confirmation" to paid; open the bloc: 1 of 3, countdown running.
5. Copy the invite link (`/products/{id}?deal={dealId}&ref={ref}`).

**Say** *I'm signed in with Google as our leader. Every listing has two prices; I start a bloc at the team price and pay first, on Stripe's own page, in Sandbox mode: a test card, no real money. Notice the receipt waits: BlocBuy believes the payment only when Stripe's signed webhook arrives, not when the browser comes back. Now the bloc exists, one of three, and I copy my invite link.*

**Expect** Receipt paid within one minute; bloc open at 1 of 3; invite link copied.

**Fallback** Checkout does not open within 20 s, or the receipt still waits after 60 s (the #517 symptom): Eric posts `CLIP 1` and says "*That is the open issue on slide 6; here is the same step recorded on dev on 25 Oct.*" No recording: Plan C (§6).

## Step 2 A friend joins through the invite and pays

16:55:20 – 16:57:20

**Proves** Consistency · **5a** (UC4, UC7) · Eric operates; Yorik narrates the guards

**Operate**

1. Window "Joiner" (account B), tab 1: paste the invite link.
2. Point at the banner that names the sharer (account A's display name).
3. **Join** → quote → Stripe Checkout → 4242 card → Pay → receipt paid.
4. Product page: the bloc now shows 2 of 3.

***Say*** *A friend opens my link. The banner names who shared it, which is how the join is credited to me. She joins and pays the same way. Behind this, the join rules run twice, at the quote and again after payment, inside the team aggregate: if the last seat had gone in between, she would get a refund obligation, never a seat that doesn't exist. Two of three.*

**Expect** Banner with the sharer's name; receipt paid; bloc at 2 of 3.

**Fallback** `CLIP 2` with the same sentence as step 1. No recording: Plan C.

## Step 3 My blocs and my orders

16:57:20 – 16:58:50

**Proves** Functionality · **5a** (UC8) · Eric

**Operate**

1. Joiner window, tab 2: `/account`. My blocs → **Active**: the bloc, 2 of 3, its countdown.
2. My orders below: the new order with its amount.
3. Click **Completed**, then **Refunded**; point at the URL (`?blocs=…`); reload: the chip survives (slide 12).

***Say*** *Her account page: My blocs on top, with Active, Completed and Refunded, and her orders below. The chip lives in the URL, so a reload, the Back button or a shared link keeps the view, and switching chips filters what is already loaded instead of asking the server again.*

**Expect** Active shows the bloc; My orders lists the new order; the chip survives a reload.

**Fallback** Page empty: reload once; then `CLIP 3`. With a NO-GO this step runs live on an account that already has orders.

## Step 4 The 16:05 bloc expires and refunds

16:58:50 – 17:01:20

**Proves** Reliability, consistency · **5a** (UC5, UC6), **5b** · Eric operates; Yorik narrates

**Operate**

1. Leader window, tab 3: `/account` → **Refunded**: the `US34-04` bloc started at ≈ 16:05, chip "Closed".
2. My orders: its order, cancelled.
3. Tab 4: its receipt `/payments/{paymentId}` (bookmarked at 16:10), reloaded: the refund status. Refund status is shown only on the receipt.
4. Never open the Stripe dashboard on the shared screen.

***Say*** *At five past four our leader started a second bloc, a team of two that lives forty minutes, and nobody joined. At about 16:45 the scheduler, which sweeps every minute, expired it. In the same transaction the order was cancelled and one refund obligation was written. A worker, every thirty seconds, sent the refund to Stripe with an idempotency key built from that obligation. Here is the receipt: refunded. We never touched the deadline; the timer did all of it.*

**Expect** Bloc under Refunded ("Closed"); order cancelled; the receipt shows the refund.

**Fallback** `REFUND LATE` at 16:50 and still no refund now: Yorik says one sentence ("the worker retries every 30 seconds; it may still be in flight"), Eric posts `CLIP 4`. No dev recording: open **B15** and say "*verified locally with a real Sandbox refund on 20 September, not yet on dev.*"

## Step 5 One seat, one refund: consistency under failure

17:01:20 – 17:02:50

**Proves** Consistency, reliability · **5b** · Yorik narrates; Eric shows

**Operate**

1. Deck tab 1, press B → **B4** (expiry and refund sequence) for about 20 s, then back to the receipt tab.
2. No live race and no webhook replay on dev: the tests below prove them.

**Say** *What you can't see is what didn't happen. Two buyers paying for the last seat at the same moment: one gets it, the other gets one refund obligation. Eight replays of the same Stripe webhook: still one member and one order. A crash after Stripe accepted a refund: the retry reuses the same key, so Stripe returns the first refund instead of making a second. Those are tests against a real PostgreSQL that run on every pull request; on dev you see the outcome, one refund per payment.*

**Expect** B4 shown briefly; the receipt with exactly one refund. Tests: StripeCheckoutFlowTest.concurrentPaidBuyersCannotBothTakeLastSeat, StripeLocalIntegrationTest.eightConcurrentWebhookReplaysStillProduceOneMemberAndOneO RefundServiceTest.lostResponseRetriesSameDurableKey.

**Fallback** If behind (§6), say only the first and last sentences.

## Step 6 Scalability and performance, labelled

17:02:50 – 17:05:20

**Proves** Scalability, performance · **5b** · Eric

**Operate**

1. Deck tab 1, press B → **B16**.
2. Left: SWE5001 prod on ECS, the scaling policy and the scale-out result.
   `[FILL: play the V3 scale-out clip (15–20 s) from the SWE5001 recording if it exists (open item 21)]`
3. Middle: the V2 k6 run on ECS (16 Oct). Right: the local workload baseline (30 Sep).

**Say** *Auto-scaling isn't on dev: dev is one host by design. The same code runs in our SWE5001 production on ECS, where commerce scales from two to six tasks on requests per target; this is that scale-out under a scripted k6 load. For performance, k6 drove one task through CloudFront from ten to two hundred requests a second: 57,600 requests, no errors, p95 under 60 milliseconds, and it never saturated. On the right, our own workload on a laptop: a p95 of 57 milliseconds for joins on one hot team. Each number says where it ran.*

**Expect** B16 with every column labelled by environment; no claim of auto-scaling on dev.

**Fallback** Clip fails: stay on B16 and read the numbers.

## Step 7 Maintainability: the pipeline you just saw

17:05:20 – 17:06:20

**Proves** Maintainability · **5b** · Eric

**Operate**

1. Leader window, tab 5: Pan's merged PR from segment 4 and the dev page with his change.

**Say** *Maintainability we showed you live a quarter of an hour ago: Pan's change went through six required checks, a code-owner review and an automatic smoke test, and it runs on the same site as this demo. Eighteen ArchUnit rules hold the module boundaries on every build, so a change in one context can't quietly reach into another.*

**Expect** The merged PR and the live change side by side.

**Fallback** PR not merged in §4: say so and show the 25 Oct recording frame instead.

**Step 8 Close** 17:06:20 – 17:07:00

| | |
|---|---|
| **Proves** | **5a**, **5b** recap · Eric, then Lim |
| **Operate** | 1. Deck tab 1: slide 21 for the recap sentence, then slide 22. |
| ***Say*** | *That's one bloc's whole life on dev: started, shared, joined, tracked, and refunded on time by the timer. Lim?* |
| **Expect** | Slide 22 at 17:07:00. |
| **Fallback** | If behind, skip straight to slide 22. |

**Never live**

Editing any deadline, price or listing on shared dev; any SSM command, port-forwarding session, database client or AWS console on the shared screen; opening the Stripe dashboard (keys and webhook secrets live there); notifications on dev (off by decision, ADR 0007: say so if asked); racing two buyers or replaying a webhook on dev; sharing the 16:05 bloc's invite link; a second payment to "try again" on air.

**Tip**

Narrate what changed, not what you click: "one of three … two of three … refunded". If a live step runs 20 seconds past its expectation, go to the clip; no troubleshooting on air.

## Closing slide

**Slide 22 Thank you** 17:07 · Q&A

| | |
|---|---|
| **Speaker** | Lim · covers guideline item **5b** |
| ***Say*** | *Thank you. We're happy to take your questions.* |
| | ≈ 8 words · ≈ 4 s at 130 wpm |

### Q&A (17:10–17:15)

| Likely question | Answer: decision → reason → evidence or limitation (30–45 s) | Owner |
|---|---|---|
| Why is there no auto-scaling on dev? | Decision: dev is one EC2 host running the monolith in Docker Compose, with PostgreSQL and Redis on the host (backend ADR 0008). Reason: SWE5006 needs one complete, always-on system at low cost; auto-scaling is the SWE5001 topology on ECS, where commerce scales 2–6 tasks on ALB requests per target (policy read back 16 Oct, B16). Limitation: the dev host is a single point of failure and a deploy restarts the API (502 on 3–4 Oct). | Eric → Pan |
| Is the refund real? | It is a real Stripe Sandbox refund: the Refund API in test mode, so no money moves. Locally on 20 Sep, refund re_3UHbeV… reached succeeded and a recheck created no duplicate (B15). On dev: `[FILL: Sandbox refund id from the 24 Oct dry run, or say "not yet proven on dev" (#223, #517)]`. The receipt reads BlocBuy's own record, which the worker writes only after Stripe answers. | Yorik |
| What happens if Stripe is down? | Nothing is admitted without a verified webhook: if Checkout cannot open, no payment exists and the quote simply expires. A missed webhook is recovered by the operator commands inspect and recover, from Stripe's own facts (20 Sep acceptance). Refunds stay as durable obligations; the worker retries every 30 s with the same idempotency key, and after 23 hours it only searches Stripe and hands mismatches to a person. Limitation: webhook recovery is an operator step, not a scheduled job. | Yorik |
| How do you avoid a double payment? | Three guards. A repeated Checkout request with the same buyer, intent and Idempotency-Key reuses the payment and its Checkout link. Webhook events are stored by event ID, so a replay changes nothing (StripeLocalIntegrationTest.eightConcurrentWebhookReplaysStillProduceOneMemberAndOneOrder). | Yorik → Eric |

| | | |
|---|---|---|
| | A purchase intent is fulfilled by one payment only; when two paid buyers race for the last seat, one gets it and the other gets a refund obligation (StripeCheckoutFlowTest.concurrentPaidBuyersCannotBothTakeLastSeat). Tests run locally and in CI, not on dev. | |
| Can you show that step again? | Yes, if it only reads: we reopen the tab (My blocs, the receipt, the pull request or the run). We do not repeat a payment or change data live; every step is also in the recordings of 25 Oct. `[FILL: recording folder link]` | Eric |
| Why did the demo not show notifications? | By decision: the dev host runs no message broker, so the outbox writer, relay and consumer are off (application-ec2.yml; ADR 0007 scope decision, 11 Oct). Notifications run on SWE5001 prod through SNS and SQS, and locally and in CI against ElasticMQ (slide 11). | Pan |
| (Questions parked during the talk) | Lim reads each parked question from the backchannel list in one line and names the owner; those come first, before new questions. Answer each in 30–45 s: decision, reason, evidence or limitation. | Lim |

# 6 Contingencies and time recovery

## If something fails

| Situation | Detected by | Response and words | Who |
|---|---|---|---|
| Presenter's audio or connection drops | Silence > 5 s | The backup continues from the slide card: Lim ↔ Yorik (§1, slide 12); Eric covers slides 8, 10 and 11, Lim covers 7 and 9 (§2); Eric ↔ Liu Jin (§3); Lim ↔ Pan (§4: slides only, the live part goes to the recording); Yorik ↔ Eric (§5: Yorik shares his standby windows). | Backup |
| Lim cannot share | Black screen | Eric shares the HTML deck (his tab 1) for segments 1–3; the PDF as a last resort. | Eric |
| Dev site down or unhealthy at 15:30 | A.1 | No changes to dev after 15:30. Segments 4 and 5 run from the 25 Oct recordings; say so on slides 6 and 21. The operator investigates only after 17:15. | Pan, Yorik |
| 16:05 bloc not confirmed | `EXPIRY LATE` at 16:10 | Retry the START once by 16:15 (expires ≈ 16:55, refund ≈ 16:57). Not paid by 16:15: step 4 uses clip 4 or B15. | Yorik |
| Refund not visible | `REFUND LATE` at 16:50 | Reload the receipt at 16:55 and before step 4. Still missing: clip 4, or B15 with "verified locally, not yet on dev". | Yorik, Eric |
| Checkout does not return, or the receipt keeps waiting (#517 symptom) | Step 1 or 2 | `CLIP 1` / `CLIP 2`; one sentence pointing to slide 6. Never retry a payment on air. | Eric |
| Demo PR checks red or still running at 16:48 | Tab 1 | Say what the gate caught; `CLIP P2` onwards; merge after 17:15 only. | Pan |
| GitHub slow or logged out | Tabs 1–4 | Recording for the remaining `P-` steps. | Pan |
| **Plan C:** a step fails live and there is no recording for it | — | Show the local evidence for that step (B15 for payment and refund, the test names in step 5, B16 for load) and say: *"verified locally, not yet on dev."* Never present a local result as a dev one. | Eric, Yorik |
| A lecturer asks for something not prepared | — | Offer to show it after the talk or point to the report; do not improvise on dev. If a backup covers it, open it (B, then the row). | Lim |

## If the clock slips

The 3-minute buffer absorbs questions during the talk. When a segment starts more than 30 s late (`BEHIND n`), cut in this order; never drop a guideline item, only depth.

| Segment | Cut first | Saves |
|---|---|---|
| 1 | The trims on slides 2–6 (each card names one) | ≈ 40 s |
| 2 | The trims on slides 7–12 (one sentence each) | ≈ 45 s |
| 3 | Slide 13: drop the PaidGroupDealTest example; slide 14: drop the ArchUnit example | ≈ 20 s |
| 4 | Slides: the trims on 15–19. Live: skip P5 (rollback) and shorten P4 to one sentence on the recorded backend run | ≈ 90 s |
| 5 | Step 3: skip the chip switching; step 5: first and last sentences only; step 7 folded into step 8 | ≈ 120 s |

# 7 Timeline before, on and after the day

## Before the day

| Date | Do | Owner |
|---|---|---|
| Mon 19 Oct | Draft deck, script and both run books (this document). **Done.** | Lim, Yorik |
| Wed 21 Oct | SWE5001 presentation (16:15–16:55). Keep its scale-out and load numbers, or the 19 Oct session results, for B16 (open item 21). | Pan, Lim |
| Fri 23 Oct | **Slides due** (#464). Each owner reviews their slides, cards and Q&A rows; the Sprint 4 review and retro feed slides 4–6. **18:00 go/no-go** for the live payment steps (§5 decision table), posted in #464. | All; Yorik decides with Lim |
| Sat 24 Oct | #223 and #517 fixed. **Full dry run on dev:** START on `US34-04` and let it expire and refund; START and JOIN on `US34-02` with both accounts; /account; payment and refund ids in the day log (no secrets). Pan rehearses P1–P3 on a throwaway PR and closes it unmerged. | Yorik (payments), Eric, Pan |
| Sun 25 Oct, by 12:00 | **Recordings** of both demos on dev, 1080p, no secrets on screen: start a `US34-04` bloc at ≈ 10:00, record steps 1–8 from ≈ 10:45; Pan records P1–P7 with a real one-line change (not Monday's). Cut one clip per step. | Yorik, Eric (system); Pan (pipeline) |
| Sun 25 Oct, afternoon | **Timed rehearsal** in Zoom at the real clock: expiry bloc at 16:05, talk at 16:30, both share switches, cue words, and lecturer questions injected during the talk. Every marker resolved after this rehearsal; rebuild deck and script. | All; Lim times |

## Presentation day (Mon 26 Oct)

| Time (SGT) | Do | Owner |
|---|---|---|
| ≤ 12:00 | A.1 health and runs; A.2 listings check; A.3 AWS read-only checks. Pan rebases the demo branch on `dev` if needed and pushes it (A.5). | Yorik, Pan |
| 14:30 | Tech check in a test Zoom: mic, camera, sharing one window (Lim, Pan) or one screen (Eric), Do Not Disturb, wired network if possible. | All |
| 15:00 | Merge freeze on `dev` until 17:15, announced in the team chat. | Lim |
| 15:30 | §4 and §5 preconditions; demo windows and tabs prepared; clips loaded. | Pan; Eric, Yorik |
| 16:00 | Join the meeting as "Team 8 – Name"; backchannel open. | All |
| ≈ 16:05 | Start the expiry bloc on `US34-04` as account A; pay; bookmark the receipt. Latest 16:15. | Yorik |
| 16:10 | `EXPIRY OK` / `EXPIRY LATE`. | Yorik |
| 16:30 | Start; follow the run sheet. | Lim |
| 16:41 | `PR OPEN`. | Pan |
| 16:46:15 | Share switch Lim → Pan. | Lim, Pan |
| 16:50 | `REFUND OK` / `REFUND LATE`. | Yorik |
| 16:53:00 | Share switch Pan → Eric. | Pan, Eric |
| 17:07 / 17:10 / 17:15 | Slide 22; parked questions; Q&A; end and thank the lecturers. | Lim |

## After the presentation

- The same evening: every lecturer remark goes into #463, each to become a report change or a justified non-change; tick the #464 acceptance criteria that are met.
- Day log under `docs/agile/sprint-04/log/`: the dev payment and refund ids shown, the demo PR and its run links (no secrets).
- The `US34-02` bloc started on air stays open for 24 h, then expires and refunds both members on its own: check it on 27 Oct as further evidence for #223.
- Lift the merge freeze; Pan merges the demo PR after 17:15 if it was not merged on air.

# 8 Open items before the 25 Oct rehearsal

There are 6 `[FILL]`, 9 `[VERIFY]` and 10 `[CONFIRM]` markers in this script (slide cards, Q&A tables and the run books; the deck shows the slide markers too). **FILL** is a value that does not exist yet, so never present a guess; **VERIFY** is a claim to check against the repository, the report or run logs; **CONFIRM** needs team agreement. All must be gone after the 25 Oct rehearsal.

| # | Item | Owner | Blocks slide |
|---|---|---|---|
| 1 | Enable Checkout, scheduling and expiry refunds on dev (#223) and fix the Stripe webhook (#517); go/no-go for the live payment demo on 23 Oct | Yorik | 21 |
| 2 | Fallback recordings of both demos on the deployed dev environment by 25 Oct | Yorik (system), Pan (pipeline) | 20, 21 |
| 3 | Sprint 1 planned and done hours: none were recorded; reconstruct from #49 or state "not tracked" | Pan | 5 |
| 4 | Sprint 3 retrospective (#451): fill the Sprint 3 → 4 row | Lim | 6 |
| 5 | Sprint 4 result, done hours, retrospective and the v0.1.0-sprint4 tag date after the 23 Oct review | Eric | 4, 5, 6 |
| 6 | Sprint 4 per-member planned hours once PR #538 merges (Liu Jin 66, Eric 26, Pan 46.5) | Eric | 5 |
| 7 | Actual effort per member, agreed by all five | Lim | 5, Q&A 1 |
| 8 | Re-check slide 6 status on 25 Oct (dev health, #517, #223, listings) | Lim | 6 |
| 9 | Confirm slide 8 (UC9 State, Optimistic Offline Lock, schema) and B5 against the §3.4–3.6 draft | Liu Jin | 8, B5 |
| 10 | Confirm slide 9 against the UC3/UC4 draft once PR #542 merges | Eric | 9 |
| 11 | Confirm slide 11 against the notifications draft; check the dev host env file leaves BLOCBUY_EVENTS_* off | Pan | 11 |
| 12 | Pick the UC8 design problem (proposal: URL as the single source of view state, one Adapter at the API boundary) | Lim | 12 |
| 13 | Confirm the testing story on slides 13–14 and B10 (no testing draft yet); recount the test numbers on 25 Oct | Liu Jin | 13, 14, B10 |
| 14 | Confirm the pipeline story on slides 15–20 and B12–B14 (no DevSecOps draft yet) | Pan | 15–20, B12–B14 |
| 15 | Choose the live demo change, its author and the approving code owner; open the PR before 16:42 | Pan, Lim | 20 |
| 16 | Dev host security group allows SSH (22) from 0.0.0.0/0 although SSM sessions work: close it or record why it stays open | Pan | 17, B14 |
| 17 | Document or remove the media CloudFront dcsjrkmummxi7 and bucket blocbuy-media-dev-402178232802 (nothing in the repository names them) | Pan | 18 |
| 18 | Reconcile frontend ADR 0004 and the EC2 operator guide with the post-merge rebuild in ci-frontend.yml and ci-backend.yml; CONTRIBUTING.md still lists contract-check as required | Pan, Lim | 15, 16, B14 |
| 19 | Fallback recording link for the pipeline demo (item 4e), part of open item 2 | Pan | 20 |
| 20 | Fallback recordings of the system demo on dev, one clip per step (item 5c), part of open item 2; link in the Q&A answer | Yorik, Eric | 21, Q&A 5 |

| | | | |
|---|---|---|---|
| 21 | SWE5001 scale-out and load results from the 19 Oct session (V3 tasks, p95, error rate; V4 drain) for B16 | Pan | B16 |
| 22 | Two Google buyer accounts (leader, joiner), their display names on screen, and the admin grant for the listings check | Yorik | 21 |
| 23 | Footer of slide 21 after the go/no-go; Sandbox payment and refund ids from the 24 Oct dry run for the Q&A | Yorik | 21, Q&A 5 |

# A Appendix — check and demo commands

Every command is read-only unless marked. Run them off the shared screen, from a checkout of `dev` at the repository root. Never paste tokens, cookie values, emails or AWS output with account details into chat or onto the shared screen.

### A.1 Health and recent runs (anyone; ≤ 12:00 and at 15:30)

```
curl -s https://d1b3np88z657cp.cloudfront.net/api/actuator/health          # expect "status":"UP"
gh run list --repo PPsyrius/iss-groupbuy-team8 --workflow ci-frontend.yml --branch dev --limit 3
gh run list --repo PPsyrius/iss-groupbuy-team8 --workflow ci-backend.yml --branch dev --limit 3
gh pr list --repo PPsyrius/iss-groupbuy-team8 --base dev --state open      # what could still merge before th
e freeze
```

### A.2 Listings check (Yorik; ≤ 12:00; the admin's blocbuy-token, typed at the prompt)

```
read -rs -p 'blocbuy-token: ' BLOCBUY_TOKEN; echo; export BLOCBUY_TOKEN
scripts/seed-dev-demo-listings.sh --check   # one quote per US34-* listing; no admin writes; non-zero on a mi
smatch
unset BLOCBUY_TOKEN
```

A policy with fewer than 7 days left needs a normal seed run (audited admin API writes) before 26 Oct; runbook `docs/deploy/backend-dev-api-agent-verification.md`. The admin grant (`scripts/grant-dev-admin.sql` over an SSM tunnel) is once per database and is done before the day by the operator, never on the day.

### A.3 AWS read-only checks (Pan; ≤ 12:00; profile `blocbuy`, Region `ap-southeast-1`)

```
aws ec2 describe-instance-status --instance-ids i-0852b6c3475b19f05 --profile blocbuy --region ap-southeast-1
aws ssm describe-instance-information --filters Key=InstanceIds,Values=i-0852b6c3475b19f05 --profile blocbuy
--region ap-southeast-1
aws cloudfront get-distribution --id E3U5W7UWDL8LRV --query 'Distribution.Status' --profile blocbuy
aws elbv2 describe-target-health --target-group-arn <dev target group arn> --profile blocbuy --region ap-sout
heast-1
```

### A.4 Payment capability (Yorik; at 15:30)

```
curl -s https://d1b3np88z657cp.cloudfront.net/api/v1/payments/capabilities
```

`[VERIFY: whether this endpoint answers without a session and reports Checkout as enabled; otherwise open it in the Leader browser]`

### A.5 Demo pull request (Pan)

**Before the day (25 Oct evening); writes to a feature branch only:**

```
git fetch origin
git switch -c feature/frontend-swe5006-live-demo origin/dev
# edit one line of user-facing copy under frontend/src (open item 15)
git commit -am "feat(frontend): <the copy change> (#464)"
git push -u origin feature/frontend-swe5006-live-demo
```

**26 Oct, ≤ 12:00**, only if `dev` moved since: `git fetch origin`, then `git rebase origin/dev`, then `git push --force-with-lease`. Never after 15:00.

**16:41, during slide 14 (Pan, muted):**

```
gh pr create --repo PPsyrius/iss-groupbuy-team8 --base dev --head feature/frontend-swe5006-live-demo \
```

```
  --title "feat(frontend): <the copy change> (#464)" --body-file <prepared body file>
```

**≈ 16:47:15, Lim on his own machine:** `gh pr review <number> --repo PPsyrius/iss-groupbuy-team8 --approve` (or the GitHub UI). **16:48, Pan:** Squash and merge in the UI, on the shared screen.

### A.6 After the talk (Yorik, Pan)

```
gh run list --repo PPsyrius/iss-groupbuy-team8 --workflow ci-frontend.yml --branch dev --limit 1   # the demo delivery run, for the day log
gh pr view <number> --repo PPsyrius/iss-groupbuy-team8 --json url,mergedAt,mergeCommit
```

# B Appendix — backup slides (Q&A only)

Backups follow slide 22 in the HTML deck and the PDF. In the deck press **B** for the index (BX) and click a row, or open `#B15` directly; slide 22 stays up otherwise. B4, B15 and B16 may also be used on air in §5.

| ID | Slide | Answers |
|---|---|---|
| B1 | Context map | How do the contexts relate, and where are the anti-corruption layers? |
| B2 | Aggregates and state machines | Which aggregates hold the rules, and how does a team end? |
| B3 | UC4 join sequence | What happens in one paid join? |
| B4 | Expiry and refund sequence | How does an expired team become refunds? |
| B5 | Database schema | Which tables does each context own? |
| B6 | UC6 candidates and decisions | What else did you consider for refunds? |
| B7 | Transition from analysis to design | How did analysis classes become design classes? |
| B8 | ADR index | Which decisions are recorded, and where? |
| B9 | Logical view (ECS topology) | How do the three services talk on ECS? |
| B10 | Test inventory | How many tests of each kind, and where do they run? |
| B11 | Sprint 2 and 3 burndowns | How did earlier sprints burn down? |
| B12 | Workflows, required checks and timings | What runs when, how long, and how often is it green? |
| B13 | Production topology (SWE5001) | What does prod look like on ECS? |
| B14 | Known gaps and limitations | What is not done yet in the pipeline and on dev? |
| B15 | Stripe Sandbox refund, local evidence (20 Sep) | Is the refund real, and where has it been proven? |
| B16 | Scale-out and load results (SWE5001 prod + dev baseline) | How does it scale, and how fast is it? |